

COLLEGE OF DEFENCE MANAGEMENT · DATATHON - 2026

Theme: Global Conflicts – Impact on Supply Chains

GLOBAL CONFLICTS AND THE BLINDING OF SUPPLY CHAINS

*A Data-Analytics Study of Chokepoint Conflict, AIS Dark Zones and Disruption
Survival, with Implications for India and the Indian Army*

Submitted by

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Accompanying files: Power BI dashboard (.pbix), star-schema data tables, analysis source code

DECLARATION

I hereby certify that:

- The work in this report is original and has been done by me for Datathon-2026 of the College of Defence Management.
- The analysis uses the unclassified datasets provided on the CDM website and open-source literature. No classified information has been used.
- Due credit has been given to all data, inferences and references from external sources, and they are cited appropriately.
- All figures, tables and statistics can be reproduced from the submitted code and data.

Place: _____

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ABSTRACT

The Datathon-2026 theme *Global Conflicts - Impact on Supply Chains* asks how conflicts affect world shipping, and specifically asks participants to identify AIS dead zones and relate them to the identification crisis created by local conflicts. This study combines eleven and a half years of weekly Middle-East conflict data (ACLED; 614,119 events) with 106,533 Sentinel-1 radar vessel detections matched to AIS during the first two weeks of the 2026 regional war (1-14 March 2026).

Change-point detection dates the regime shifts in regional violence to 7 October 2023 and 28 February 2026 without being given either date. The war regime ran at 2.9 times the previous year's rate and was dominated (84 per cent) by stand-off strikes, which reached the GCC energy coast (124-fold rise). A new Chokepoint Conflict Intensity Index shows that the Red Sea threat moved offshore and that the Hormuz littoral reached 188 times its baseline. Survival analysis of 32 disruption episodes shows a heavy tail: 62% outlast India's strategic petroleum reserve.

Large ships were AIS-dark in 74% of cases in the Strait of Hormuz and 38% in the Black Sea, against about 10 per cent in peacetime waters (relative risk 4.7, odds ratio 8.0). Getis-Ord hot spots and DBSCAN clusters locate fleets of dark tankers waiting at the Dubai and Fujairah anchorages. A gradient-boosted model validated on unseen sea regions (ROC-AUC 0.69) predicts AIS dark spots from conflict geography.

The report concludes that the first measurable effect of war at a chokepoint is the *blinding* of the supply chain, followed by physical disruption that lasts longer than national buffers. It assesses the impact on the globe, India and the Indian Army, gives a prescriptive stock-cover model (returns flatten at about 35 days of cover), and makes twelve recommendations, a phased way forward and a Power BI decision dashboard.

Keywords: supply chain, chokepoints, AIS dark vessels, SAR, ACLED, change-point detection, Getis-Ord Gi*, DBSCAN, survival analysis, spatial cross-validation, Indian Army logistics.

LIST OF ABBREVIATIONS

ACLED	Armed Conflict Location & Event Data
AIS	Automatic Identification System
AUC	Area Under the (ROC) Curve
CCII	Chokepoint Conflict Intensity Index
CI	Confidence Interval
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
FoC	Flag of Convenience
GBT	Gradient-Boosted Trees
GCC	Gulf Cooperation Council
GNSS	Global Navigation Satellite System
HR	Hazard Ratio
IFC-IOR	Information Fusion Centre - Indian Ocean Region
IMEC	India-Middle East-Europe Economic Corridor
INSTC	International North-South Transport Corridor
KM	Kaplan-Meier
MDA	Maritime Domain Awareness
MID	Maritime Identification Digits
MMSI	Maritime Mobile Service Identity
OR	Odds Ratio
PELT	Pruned Exact Linear Time
POL	Petroleum, Oil and Lubricants
RR	Relative Risk
SAR	Synthetic Aperture Radar
SPR	Strategic Petroleum Reserve
VLCC	Very Large Crude Carrier
WWR	War Wastage Reserve

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1. INTRODUCTION

Around 80 per cent of world merchandise trade by volume moves by sea, and a large share of it passes through a handful of narrow chokepoints: the Strait of Hormuz, Bab-el-Mandeb, the Suez Canal, the Strait of Malacca, the Bosphorus and a few others [1, 2]. When conflict reaches one of these straits, the effect spreads through energy prices, freight rates, insurance premia and delivery lead-times to every economy that depends on them. Between October 2023 and June 2026 the Middle East produced two such shocks in quick succession. The first was the Houthi anti-shipping campaign in the Red Sea and Gulf of Aden, which diverted a large part of Asia-Europe container traffic around the Cape of Good Hope [1]. The second was the regional war that broke out on 28 February 2026 and carried missile and drone strikes to the Iranian coast and the Gulf Cooperation Council (GCC) states around the Strait of Hormuz.

Conflict affects shipping in two ways. The obvious effect is physical: ships are struck, diverted or held in port. The less obvious effect is **informational**. Ships in or near a war zone switch off or manipulate their Automatic Identification System (AIS) transponders to avoid being targeted. Electronic warfare jams and spoofs satellite navigation. Sanctions-evading fleets operate dark by design. The common maritime picture, which navies, insurers, port authorities and logisticians all rely on, then loses its identities. The Datathon-2026 problem statement calls this an *identification crisis* and asks participants to find AIS dead zones and relate them to local conflicts such as those in the Black Sea and the Persian Gulf [3].

This report takes up that problem with two unclassified datasets from the Datathon repository and a set of descriptive, diagnostic, predictive and prescriptive techniques. The first dataset is eleven and a half years of weekly Middle-East conflict events (ACLED). The second is 1-14 March 2026 of Sentinel-1 synthetic-aperture-radar (SAR) vessel detections matched against AIS. The report works from the evidence to its impact on the globe, on India and on the Indian Army, and ends with recommendations, a way forward and conclusions that are each traced back to a finding in the analysis.

Thesis. Conflict near a maritime chokepoint shows up in the data first as the *blinding* of the supply chain: large, AIS-obligated ships vanish from the cooperative picture. Only after that does it show up as the physical disruption of trade flows. In March 2026, large ships in the Gulf war zone were 4.7 times more likely to be AIS-dark than large ships elsewhere, and distance from the fighting predicts darkness even in regions the model has never seen. Disruption episodes have a heavy tail: about 62% outlast the ~9.5 days of India's strategic petroleum reserve. India's economic and military exposure therefore has to be managed as one data problem, joining maritime domain awareness with stock-holding policy.

1.1 Problem Statement

Can open conflict and satellite data be used to (i) measure how Middle-East conflict intensity near maritime chokepoints has changed, (ii) locate and explain AIS dark zones and relate them to that conflict, (iii) predict where AIS darkness will appear, and (iv) turn the results into quantified guidance for India's supply-chain security and for the Indian Army's logistics and operational preparedness?

1.2 Aim

To derive data-driven insights on the impact of global, and in particular Middle-East, conflicts on maritime supply chains, with a focus on AIS dark zones and chokepoint disruption, and to recommend measures for India and the Indian Army.

1.3 Objectives

- a. Build a clean, reproducible data pipeline for the ACLED conflict and Sentinel-1 SAR/AIS datasets (Chapter 4).
- b. Characterise the Middle-East conflict system from 2015 to 2026, detect statistically significant regime shifts and measure the spill-over of the 2026 war onto the GCC energy coast (Chapter 5).
- c. Construct a Chokepoint Conflict Intensity Index (CCII), forecast it and model how long disruption episodes last with survival analysis (Chapter 6).
- d. Map AIS darkness worldwide, test whether it is significantly higher in war zones, find statistically significant dark hot spots and dark-vessel clusters, and profile the identity and registry of ships in the war zone (Chapter 7).
- e. Train and validate a model that predicts AIS dark spots from conflict geography (Chapter 8).
- f. Turn the findings into impact assessments for the globe, India and the Indian Army, plus a prescriptive stock-cover model, recommendations and a way forward (Chapters 9-12).

1.4 Hypotheses

Table 1.1 : Hypotheses tested in this study

#	Hypothesis	Test
H1	Middle-East political violence underwent statistically significant regime shifts that coincide with the 2023 and 2026 wars.	Change-point detection (PELT), Mann-Whitney U
H2	The 2026 war moved conflict onto the maritime/energy littoral (GCC, Hormuz) and the Red Sea threat moved offshore.	Littoral and at-sea indices, rate multipliers
H3	Large (≥ 100 m) vessels are significantly more often AIS-dark in war zones than elsewhere.	Chi-square, odds ratio, Wilson intervals
H4	AIS darkness clusters in statistically significant hot spots that coincide with conflict chokepoints.	Getis-Ord G_i^* , DBSCAN
H5	Proximity to conflict predicts AIS darkness out-of-region.	Logit, gradient-boosted trees, leave-one-region-out CV
H6	Checkpoint disruption durations are heavy-tailed and often exceed India's strategic oil cover.	Kaplan-Meier, Cox PH, expected shortfall

1.5 Methodology Overview

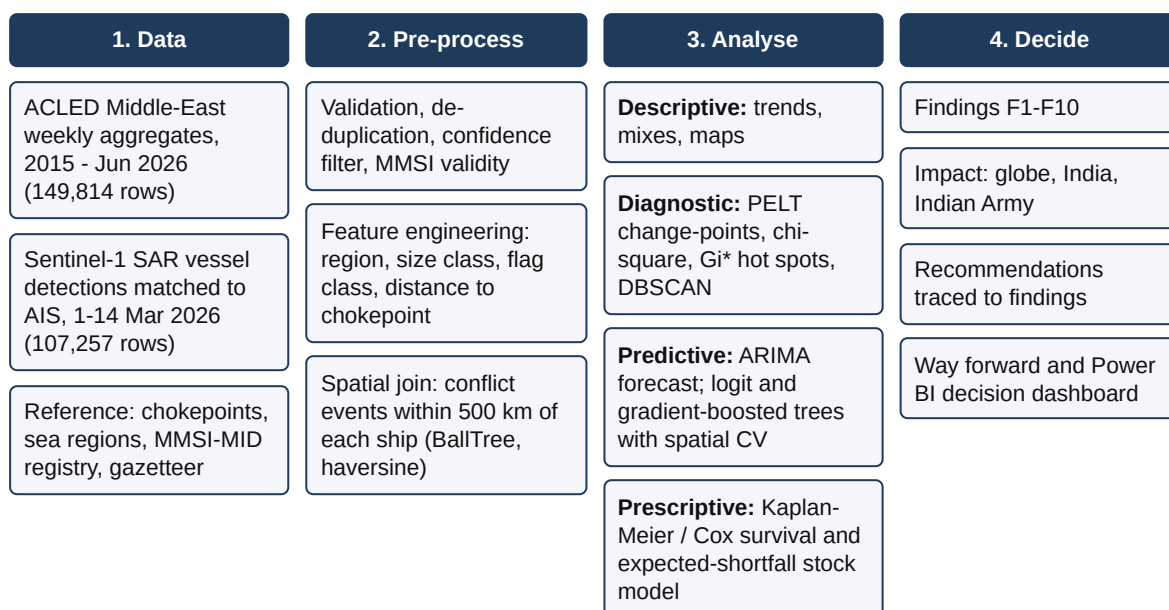


Figure 1.1 : Analytical framework - from data to decision

The work was done in Python 3.11 (pandas, NumPy, SciPy, statsmodels, scikit-learn, ruptures, Matplotlib/Basemap). The cleaned star-schema tables are exported for the Power BI (.pbix) dashboard that accompanies the report (Appendix B). Running the whole pipeline with a single command (*analysis/run_all.py*) regenerates every figure, table and number in this report from the raw files.

1.6 Organisation of the Report

Chapter 2 reviews the literature and Chapter 3 sets out the theory behind the techniques used. Chapter 4 covers data pre-processing. Chapters 5 to 8 present the analysis: the conflict landscape, chokepoint intensity and disruption survival, AIS dark zones, and the predictive model. Chapter 9 brings the findings together and assesses their impact on the globe, India and the Indian Army. Chapters 10, 11 and 12 give the recommendations, the way forward and the conclusion, followed by the references and appendices.

2. LITERATURE REVIEW

The study drew on work in conflict-event data, maritime remote sensing, maritime security and statistical methods. The works that shaped the study most are summarised below.

- a. **Raleigh et al. (2010)** introduced the Armed Conflict Location and Event Data (ACLED) project, which codes political violence and protest events by date, location, actor and type [4]. The event typology used here (battles, explosions/remote violence, violence against civilians, riots, protests, strategic developments) and the weekly ADMIN1 aggregates come from ACLED. Aggregation to province centroids limits spatial precision, so this study measures distance to conflict at ADMIN1 resolution and treats the at-sea regions separately.
- b. **Paolo et al. (2024)**, writing in *Nature*, combined Sentinel-1 SAR imagery with deep learning and AIS to map industrial activity at sea. They found that about 72-76 per cent of the world's industrial fishing vessels and about a quarter of transport and energy vessels are not publicly tracked [5]. The SAR detections, presence, length and fishing scores and the AIS-matching logic in the Datathon SAR dataset follow the same approach. That result gives a *baseline* level of darkness against which conflict-related darkness can be measured.
- c. **UNCTAD, Review of Maritime Transport (2024)** documented the Red Sea crisis: collapsing Suez and Bab-el-Mandeb transits, Cape of Good Hope diversions adding roughly 10-14 days to Asia-Europe voyages, and the knock-on effects on freight rates, emissions and developing-country trade costs [1]. It frames the physical-disruption channel that this study pairs with the informational (AIS) channel.
- d. **U.S. Energy Information Administration (2024)**, *World Oil Transit Chokepoints*, estimates that flows through the Strait of Hormuz are about one-fifth of global petroleum liquids consumption, with most of them going to Asian buyers including India, and that alternative pipeline bypass capacity is limited [2]. This makes Hormuz the single most consequential chokepoint for India.
- e. **Killick, Fearnhead and Eckley (2012)** proposed PELT (Pruned Exact Linear Time), an exact change-point search with linear cost [6]. **Truong, Oudre and Vayatis (2020)** reviewed offline change-point methods and released the *ruptures* library used here [7].
- f. **Getis and Ord (1992) and Ord and Getis (1995)** developed the G and Gi* local statistics for finding significant spatial clusters of high or low values (hot and cold spots) [8, 9]. **Ester et al. (1996)** proposed DBSCAN, a density-based clustering algorithm that finds arbitrarily shaped clusters and treats sparse points as noise [10]. It suits the detection of anchorages and ship-to-ship transfer areas.
- g. **Kaplan and Meier (1958) and Cox (1972)** established the non-parametric survival estimator and the proportional-hazards model [11, 12]. The author's earlier work on

predictive maintenance of armoured vehicles applied the same techniques to equipment time-to-failure data [13]. Here they are transferred to the time-to-end of chokepoint disruption episodes.

- h. **Roberts et al. (2017)** showed that ordinary random cross-validation overstates model skill when data are spatially autocorrelated and recommended blocked (e.g. leave-one-region-out) validation [14]. **Friedman (2001)** introduced gradient boosting [15]. **Hyndman and Athanasopoulos (2021)** give the forecasting practice (ARIMA, exponential smoothing, hold-out back-testing) followed here [16].

Gap addressed. The literature treats conflict data, SAR/AIS data and supply-chain economics separately. No study found joins weekly conflict intensity at chokepoints to SAR-observed AIS darkness during a live war. Nor has any study turned the resulting disruption-duration distribution into a stock-holding decision for an Indian user. That join is what this study contributes.

3. BACKGROUND

3.1 Maritime Chokepoints and Supply Chains

A chokepoint is a narrow channel on a widely used sea route that cannot easily be bypassed. Closing or degrading one adds distance, time and insurance cost to every voyage that depends on it. For some cargoes (Gulf crude, LPG and LNG, Qatari gas, Gulf fertiliser) there is little or no alternative supply in the short run. The Strait of Hormuz (~39 km wide at its narrowest point) connects the Persian Gulf producers to the Arabian Sea. Bab-el-Mandeb (~29 km) connects the Gulf of Aden to the Red Sea and the Suez Canal. Both lie within 2,000-3,000 km of India's west coast.

3.2 AIS, SAR and the Identification Crisis

AIS is a VHF transponder system that broadcasts a ship's identity (MMSI), position, course and speed. Under SOLAS Chapter V, Regulation 19, AIS is mandatory for ships of 300 gross tonnage and above on international voyages, cargo ships of 500 GT and above, and all passenger ships [17]. The first three digits of the MMSI, the Maritime Identification Digits (MID), identify the flag state. AIS is cooperative: a ship can switch it off, transmit a false identity, or be displaced by GNSS spoofing so that its reported position no longer matches its real one.

SAR satellites such as Sentinel-1 image the sea surface day and night through cloud. A ship shows up as a bright return whatever it chooses to transmit. Matching SAR detections to AIS tracks at the image time separates *cooperative* (matched) ships from *non-cooperative* or *dark* (unmatched) ones. Small craft are dark almost everywhere because they are not required to carry AIS. A dark ship over 100 m long, which is almost certainly AIS-obligated, is an anomaly. This study uses the large-ship dark share as its main indicator of the identification crisis.

3.3 Analytical Techniques

3.3.1 Proportions, confidence intervals and association

The share of dark ships in a region is a binomial proportion $p = x / n$. Its 95 per cent Wilson score interval [18] is

$$\left(p + z^2/2n \pm z \sqrt{p(1-p)/n + z^2/4n^2} \right) / \left(1 + z^2/n \right), \quad z = 1.96 \quad (3.1)$$

Association between war-zone location and darkness is tested with Pearson's chi-square on the 2x2 contingency table, and its strength is expressed as the relative risk $RR = p_{\text{war}} / p_{\text{rest}}$ and the odds ratio $OR = (a \cdot d) / (b \cdot c)$. Differences in weekly event counts between periods are

tested with the non-parametric Mann-Whitney U test, which assumes no particular distribution.

3.3.2 Change-point detection (PELT)

For a weekly series $y_{1..T}$, PELT finds the change-points $\tau_1 < \dots < \tau_m$ that minimise

$$\sum_{i=0..m} C(y_{\tau_i+1 : \tau_{i+1}}) + \beta m, \quad C = \sum (y_t - \bar{y})^2 \text{ (L2 cost)} \quad (3.2)$$

where β is a penalty that prevents over-segmentation. The series was standardised, a penalty of 12 and a minimum segment of three weeks were used, and the search was exact (jump = 1).

3.3.3 Local spatial statistics: Getis-Ord G_i^*

For grid cell i with value x_i (dark share) and binary neighbour weights w_{ij} (the cell itself and its eight neighbours):

$$G_i^* = (\sum_j w_{ij} x_j - \bar{X} \sum_j w_{ij}) / (S \sqrt{[n \sum w_{ij}^2 - (\sum w_{ij})^2] / (n-1)}) \quad (3.3)$$

G_i^* is a z-score. Values above 1.96 mark a statistically significant hot spot at the 5 per cent level.

3.3.4 Density clustering (DBSCAN)

DBSCAN groups points that have at least $minPts$ neighbours within radius ϵ . Using the haversine metric, $\epsilon = 25$ km and $minPts = 15$ on dark ships of 100 m and over, it finds anchorages and loitering or transfer areas where large ships gather with their AIS off.

3.3.5 Forecasting

The weekly Hormuz index is log-transformed ($y' = \ln(1+y)$) and modelled as ARIMA(1,0,1) with a constant:

$$y'_t = c + \phi y'_{t-1} + \theta \epsilon_{t-1} + \epsilon_t \quad (3.4)$$

It is benchmarked against a naive forecast and damped-trend exponential smoothing on a 12-week hold-out.

3.3.6 Survival analysis

A *disruption episode* is a run of weeks in which a theatre's index is above its threshold (baseline mean + 3 SD). Its duration T is the survival time, and an episode still under way at the end of the data is right-censored. The Kaplan-Meier estimator of $S(t) = P(T > t)$ is

$$\hat{S}(t) = \prod_{t_i \leq t} (1 - d_i / n_i) \quad (3.5)$$

where d_i episodes end at t_i out of n_i still at risk. Covariate effects are estimated with the Cox proportional-hazards model $h(t|x) = h_0(t) \exp(\beta'x)$, and theatres are compared with the log-rank test.

3.3.7 Classification and spatial cross-validation

The probability that a detected ship is dark is modelled with logistic regression, $\text{logit}(p) = \beta_0 + \beta'x$, which gives interpretable odds ratios, and with histogram gradient-boosted trees with monotonic constraints, which capture non-linear effects. Skill is measured by leave-one-region-out cross-validation, so every prediction is made for a sea region the model did not see in training [14], and reported as ROC-AUC, PR-AUC and Brier score.

3.3.8 Expected shortfall of a stock policy

If a disruption lasts T days and stock covers S days, the uncovered days are $(T - S)^+$. With the Kaplan-Meier curve,

$$E[(T - S)^+] = \int_S^\infty \hat{S}(t) dt \quad (3.6)$$

This turns the disruption-duration distribution into a stock-holding decision curve (Section 9.4).

4. DATA PREPROCESSING

Pre-processing is the backbone of any statistical model. Both datasets were loaded into pandas, validated, cleaned, enriched with derived features and written to a clean columnar store (Parquet). The two datasets are summarised below.

Table 4.1 : Summary of the conflict dataset (ACLED Middle-East weekly aggregates)

Parameter	Value
Study period	03 Jan 2015 - 27 Jun 2026
Weeks covered	600
Countries / sea areas	17
ADMIN1 units	243
Event-type classes	6
Sub-event classes	25
Total events	614,119
Total fatalities	552,700
Rows (country-admin1-week-subevent)	149,814

Table 4.2 : Summary of the maritime dataset (Sentinel-1 SAR detections matched to AIS)

Parameter	Value
Observation window	01 Mar 2026 - 14 Mar 2026
Sentinel-1 scenes	2,833
Vessel detections	106,533
Detections matched to AIS	64,543 (60.6%)
Detections with no AIS match ('dark')	41,990 (39.4%)
Median vessel length (m)	64
Large vessels (≥ 100 m)	40,179
Distinct MMSIs	53,479
Sea regions defined	17

Note on coverage. Although the SAR file is named *indian ocean vessel*, its detections cover the world (latitude -56° to 75°), with 2,833 Sentinel-1A scenes. This allows a controlled comparison: war zones (Gulf, Black Sea) against India's seas and the rest of the world, all imaged by the same sensor in the same fortnight.

Table 4.3 : Data headings analysed

Attribute	Dataset	Description
scene_id	SAR	Sentinel-1A scene identifier (image footprint)
timestamp	SAR	Image acquisition time (UTC)
lat / lon	SAR	Detected hull position
presence_score	SAR	CNN confidence that the object is a vessel (0-1)
length_m	SAR	Estimated hull length from SAR backscatter
mmsi	SAR	AIS identity of the matched track (blank when none)
matching_score	SAR	Confidence of the SAR-AIS association
fishing_score	SAR	Model probability that the hull is a fishing vessel
matched_category	SAR	AIS vessel class, or 'unmatched' (AIS-dark)
WEEK	ACLED	Week (Saturday start) of aggregation
COUNTRY / ADMIN1	ACLED	Location of events (incl. 'Indian Ocean', 'Mediterranean Sea')
EVENT_TYPE / SUB_EVENT_TYPE	ACLED	6 event types, 25 sub-event types
EVENTS / FATALITIES	ACLED	Counts per week-location-type
POPULATION_EXPOSURE	ACLED	Population within reach of events
CENTROID_LAT/LON	ACLED	ADMIN1 centroid used for geo-joins

4.1 Cleaning

Table 4.4 : Data-cleaning log

Dataset	Step	Count
ACLED	Rows loaded	149,825
ACLED	Exact duplicate rows removed	0
ACLED	Pre-2015 partial-coverage rows dropped	11
ACLED	Negative counts found	0
ACLED	Population exposure NaNs imputed (ADMIN1 median, else 0)	30,965
ACLED	Rows after cleaning	149,814
SAR	Rows loaded	107,257
SAR	Duplicate detections (same scene & position) removed	0
SAR	Invalid coordinates	0
SAR	Low-confidence detections dropped (presence < 0.80)	724
SAR	Malformed MMSI values flagged (outside 2xx-7xx MID range)	1,565
SAR	Rows after cleaning	106,533

- **Temporal integrity.** One partial week (27 Dec 2014) was dropped so that every year is complete. Weeks were re-indexed to a regular 7-day frequency, with zero-filling, before any time-series operation.
- **Missing values.** POPULATION_EXPOSURE is missing in 30,965 rows, mostly sea areas and sparsely populated provinces. It was imputed with the median of the same

ADMIN1 unit, or 0 where the unit never reports it. No other field had missing values.

- **Confidence filter.** 724 SAR detections with presence score < 0.80 were dropped to remove sea clutter, wakes and ambiguous returns.
- **Identity validity.** 1,565 MMSIs outside the ship-station range (MID 2xx-7xx) were flagged as malformed and kept for the identity-integrity analysis (Section 7.7).
- **Definition of darkness.** *Dark* = matched_category 'unmatched' (no confident AIS association). This covers 39.4% of detections. A stricter definition (no candidate MMSI at all, 26.8%) was used as a robustness check and gives the same regional ranking.

4.2 Feature Engineering

- **Sea region:** 17 analyst-defined boxes (Strait of Hormuz, Persian Gulf, Gulf of Oman, Red Sea, Bab-el-Mandeb, Gulf of Aden, Arabian Sea, Bay of Bengal, Strait of Malacca, etc.), grouped into zones: Gulf war zone, Black Sea war zone, Red Sea zone, India's seas, rest of world.
- **Size class** from SAR length (<25 , 25-40, 40-100, 100-200, >200 m). The *large* flag (≥ 100 m) marks ships that almost certainly carry AIS.
- **Flag state** decoded from the MID and classed as national flag, flag of convenience (ITF list) or shadow-fleet-associated registry.
- **Distance to the nearest chokepoint** (haversine) for every ship and every ADMIN1 centroid.
- **Conflict exposure** of every ship: distance to the nearest ADMIN1 with political violence in the war weeks overlapping the SAR window, and the number of such events within 500 km (BallTree haversine queries). At-sea ACLED rows carry a nominal basin centroid rather than a real position, so they were excluded from this spatial join.
- **Conflict flags:** political violence, remote violence (air/drone/missile/artillery), stand-off strike, GCC, maritime.

5. CONFLICT LANDSCAPE OF THE MIDDLE EAST, 2015-2026

The ACLED data record 614,119 events and 552,700 reported fatalities across 243 provinces in 600 weeks. This chapter describes how the conflict system evolved and tests hypotheses H1 and H2.

Middle-East conflict volume, 2015-2026

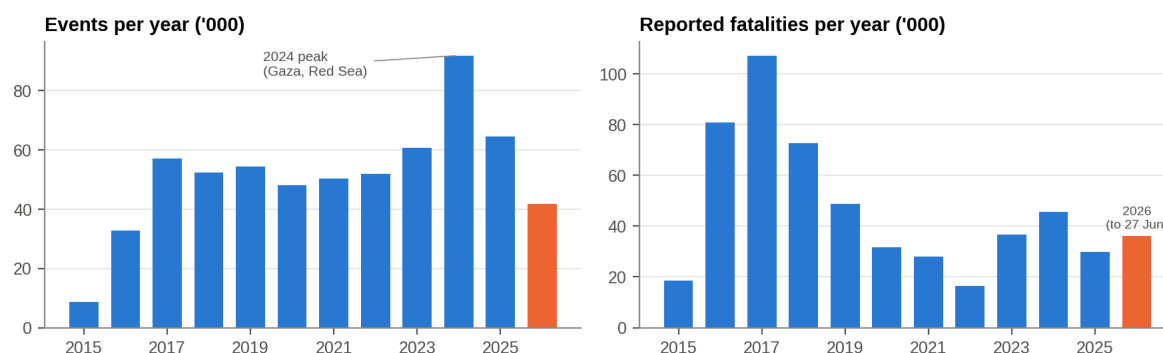


Figure 5.1 : Annual events and reported fatalities, 2015-2026 (2026 = 26 weeks)

Event volume peaked in 2024 (91,835 events), driven by the Gaza war, the Israel-Hezbollah escalation and the Red Sea campaign. In only 26 weeks, 2026 has already reached 41,865 events, and 2026 fatalities are the highest since 2019. Part of that fatality count comes from ACLED's coding of the January 2026 unrest in Iran: 27,496 fatalities in the single week of 3 January 2026, almost all coded as riots and protests. The rest comes from the war that began on 28 February.

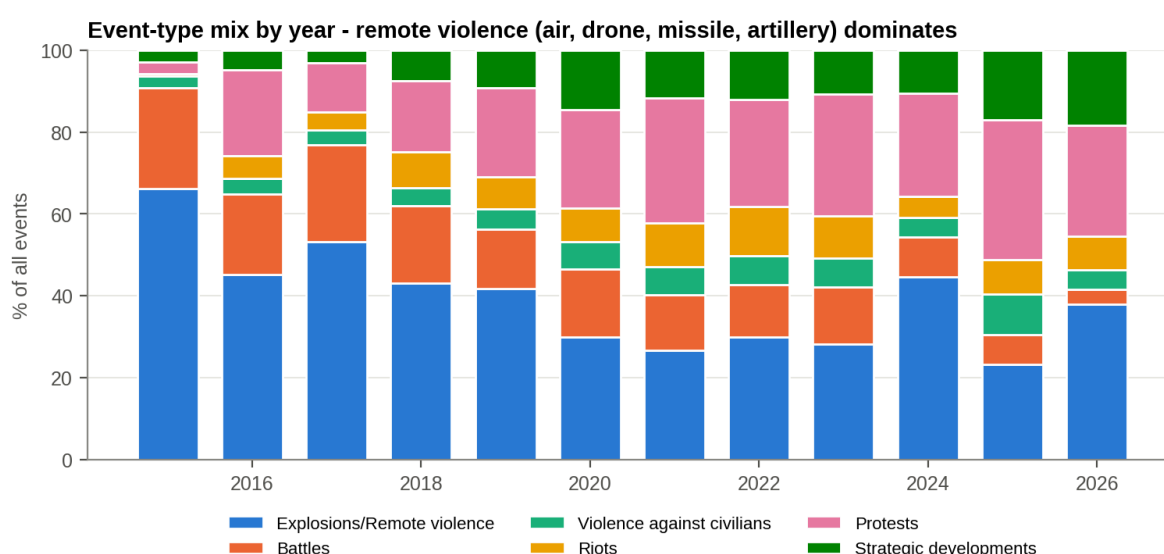


Figure 5.2 : Share of event types by year

Explosions/remote violence (air and drone strikes, shelling, missiles, IEDs) is the largest event type in most years. Its share rises sharply in years of inter-state war: 44.7 per cent in

2024 and 38.0 per cent in 2026. In between, protests and strategic developments dominate. The pattern for 2026 is clear: stand-off strikes (air/drone strike plus shelling/missile) made up **84 per cent** of all political violence recorded between 28 February and 10 April 2026.

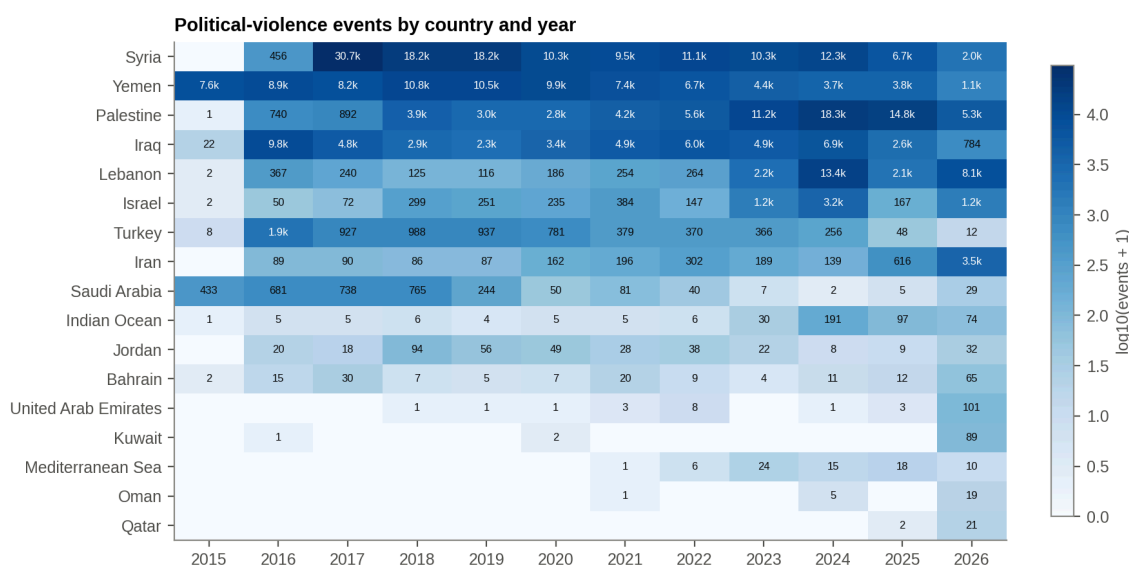


Figure 5.3 : Political-violence events by country and year (log colour scale)

The heat map shows the conflict moving across the region. Syria and Iraq dominated from 2015 to 2018. Palestine, Israel and Lebanon rose after 2023. In 2026 Iran and the GCC states, which had previously seen little political violence, turned dark. Bahrain, Kuwait, the UAE, Qatar and Oman record more political violence in the first half of 2026 than in the previous decade.

5.1 Regime Shifts (H1)

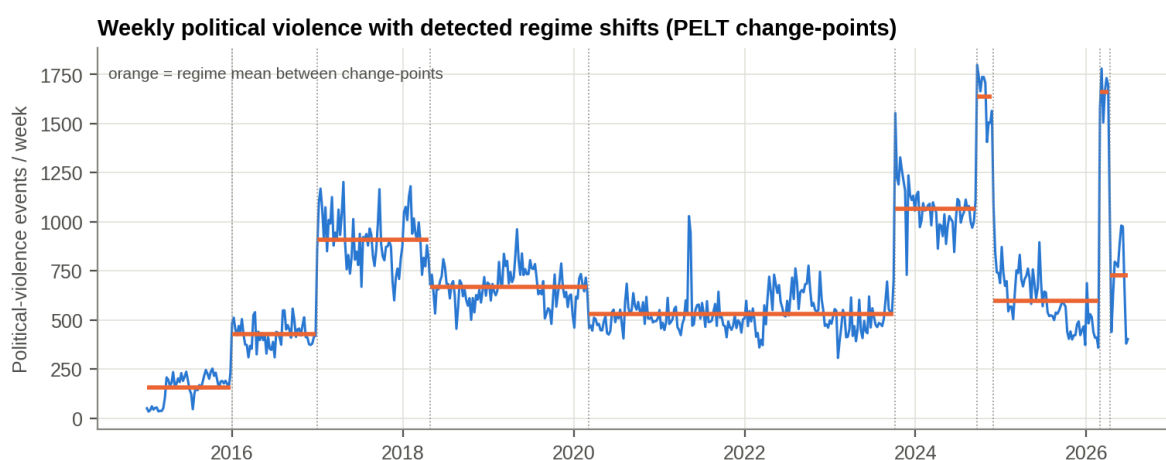


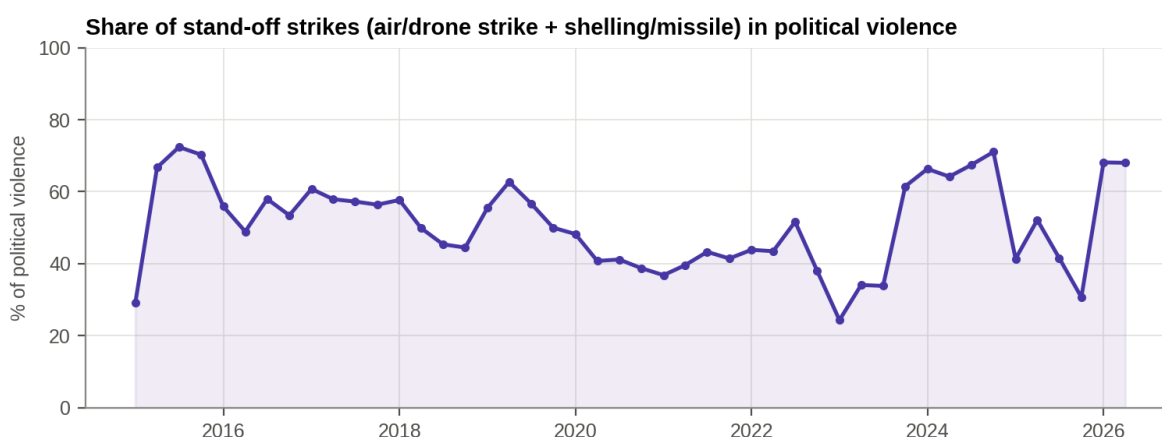
Figure 5.4 : Weekly political violence with PELT change-points (orange = regime mean)

Table 5.1 : Conflict regimes detected by PELT

From	To	Weeks	Mean events/wk	SD
03 Jan 2015	26 Dec 2015	52	155	69
02 Jan 2016	24 Dec 2016	52	427	62
31 Dec 2016	21 Apr 2018	69	909	131
28 Apr 2018	29 Feb 2020	97	667	90
07 Mar 2020	30 Sep 2023	187	531	95
07 Oct 2023	14 Sep 2024	50	1,067	129
21 Sep 2024	23 Nov 2024	10	1,635	131
30 Nov 2024	21 Feb 2026	65	597	145
28 Feb 2026	04 Apr 2026	6	1,658	102
11 Apr 2026	27 Jun 2026	12	727	230

Without being told any dates, PELT placed change-points on **7 October 2023** (the Hamas attack on Israel and the start of the Gaza war), **21 September 2024** (the Israel-Hezbollah escalation), **30 November 2024** (the collapse of the Assad government followed shortly after) and **28 February 2026** (the regional war), with a de-escalation break on **11 April 2026**. Weekly political violence averaged 1,658 events in the war regime against 568 in the preceding 52 weeks, a 2.9-fold rise (Mann-Whitney U, $p = 3.6e-05$). After the April break, violence settled at about 727 events a week, still above the pre-war level. **H1 is supported.**

5.2 Stand-off Warfare and the Spill-over to the Energy Coast (H2)

**Figure 5.5 : Share of stand-off strikes (air/drone strike, shelling/missile) in political violence, quarterly**

Stand-off strikes are not trending steadily upwards. Their share *jumps* when an inter-state war begins (linear trend $p = 0.32$), reaching 68 per cent in Q1-2026. The share of stand-off strikes therefore works as a signature of war: a rise in it is an early-warning signal that ranges, and with them threats to shipping and energy infrastructure, are about to extend.

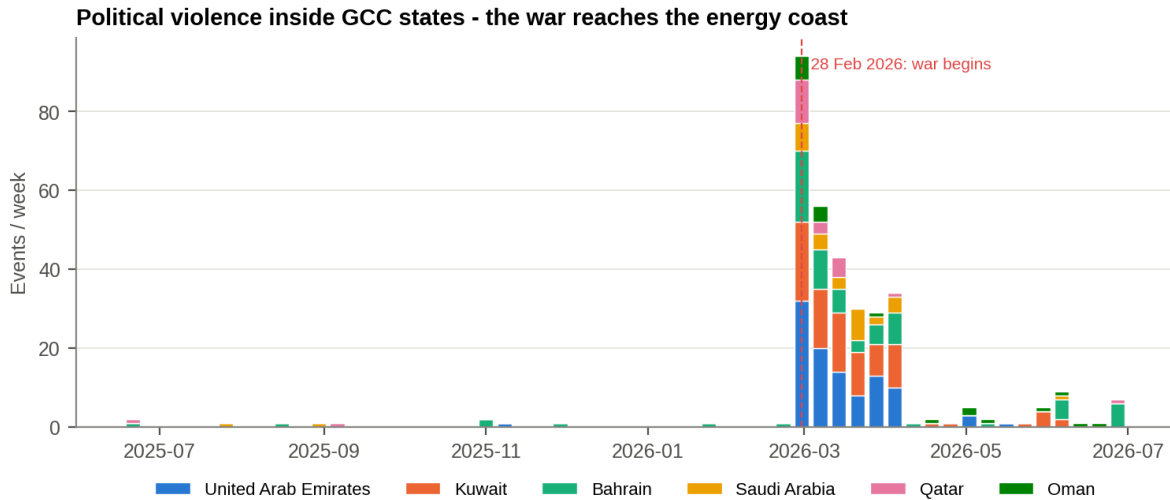


Figure 5.6 : Weekly political violence inside the GCC states, June 2025 - June 2026

Before the war the six GCC states together recorded about 0.4 political-violence events a week. In the six weeks from 28 February 2026 this rose to 47.7 a week, a **124-fold** increase. The UAE, Kuwait, Bahrain and Qatar, which host the region's export terminals, refineries and LNG trains, were struck directly.

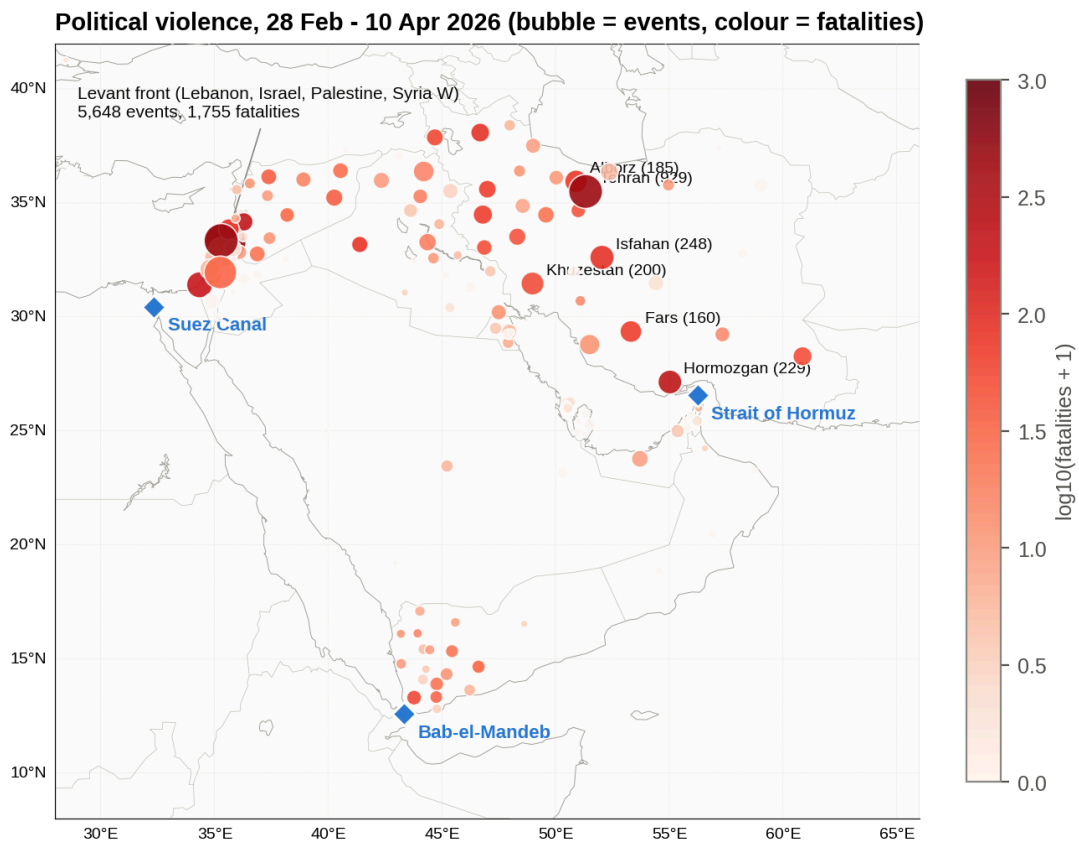


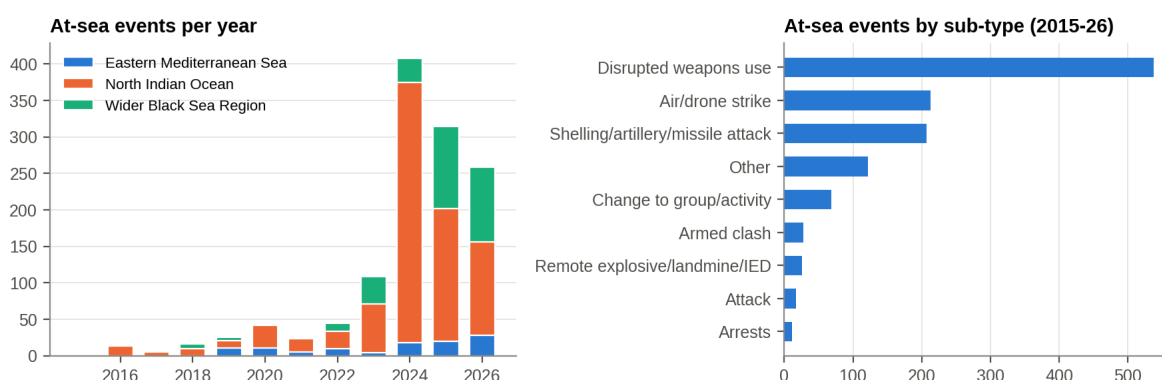
Figure 5.7 : Geography of political violence, 28 February - 10 April 2026

Table 5.2 : Top provinces by political-violence events, 28 Feb - 10 Apr 2026

COUNTRY	ADMIN1	EVENTS	FATALITIES
Lebanon	Al Nabatieh	1,990	549
Lebanon	South	968	463
Iran	Tehran	929	466
Palestine	West Bank	800	32
Israel	HaZafon	586	3
Palestine	Gaza Strip	328	180
Iran	Isfahan	248	90
Iran	Hormozgan	229	215
Iran	Khuzestan	200	51
Iran	Alborz	185	76
Iran	Fars	160	68
Iraq	Erbil	133	15
Lebanon	Mount Lebanon	129	83
Iran	Bushehr	125	11
Israel	Tel Aviv	111	5

Hormozgan, the Iranian province on the northern shore of the Strait of Hormuz, recorded 229 events and 215 fatalities in six weeks. Bushehr, Khuzestan (the oil province at the head of the Gulf) and Fars were also among the most affected Iranian provinces. The fighting reached the shoreline of the world's most important energy chokepoint.

Conflict at sea - the Red Sea/Indian Ocean campaign

**Figure 5.8 : Conflict at sea: ACLED at-sea events by year and by sub-type**

ACLED recorded 44 at-sea events in 2022, 108 in 2023, 408 in 2024 and 314 in 2025, and already 258 in the first half of 2026. The dominant sub-types, 'disrupted weapons use' (missiles and drones intercepted over the sea) and shelling/missile attacks, are those of an anti-shipping campaign. **H2 is supported** (see also Section 6.1).

6. CHOKEPOINT CONFLICT INTENSITY AND DISRUPTION SURVIVAL

6.1 The Chokepoint Conflict Intensity Index (CCII)

Each theatre's threat to shipping is measured where it appears in the data. Weekly political-violence events are weighted 1.5 for stand-off strikes (the type of attack that reaches ships and terminals) and 1.0 otherwise:

- **Hormuz (littoral):** events in ADMIN1 units within 500 km of the strait (coastal Iran, UAE, Oman, Qatar).
- **Red Sea / Arabian Sea (at sea):** ACLED's at-sea 'North Indian Ocean' events, i.e. the Houthi anti-shipping campaign.
- **East Med (at sea) and Black Sea (at sea):** ACLED's at-sea events for those waters.

A theatre is *disrupted* in a week when its index is above its own threshold, set at the 2015-2022 baseline mean plus 3 SD (minimum 3 weighted events).

Chokepoint Conflict Intensity Index (weekly), 2015-2026

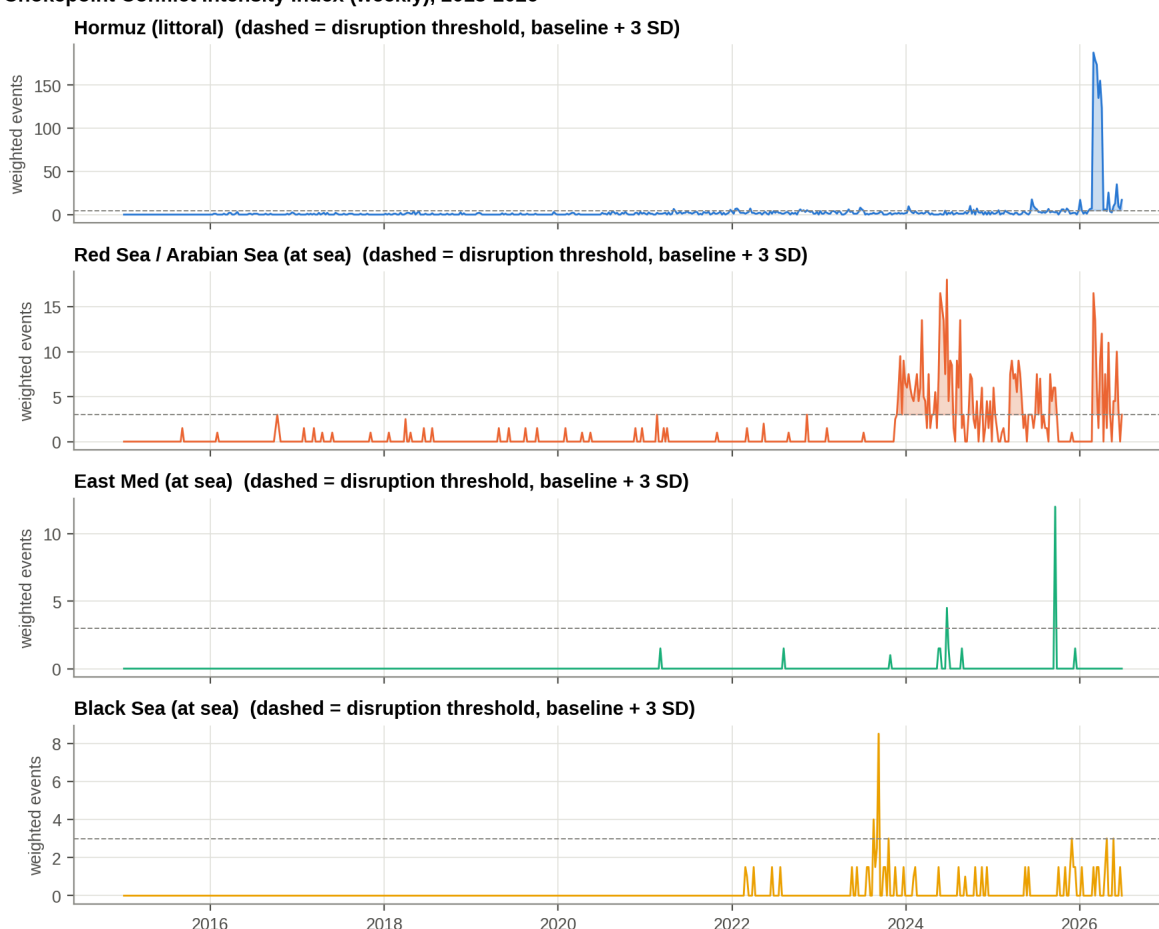
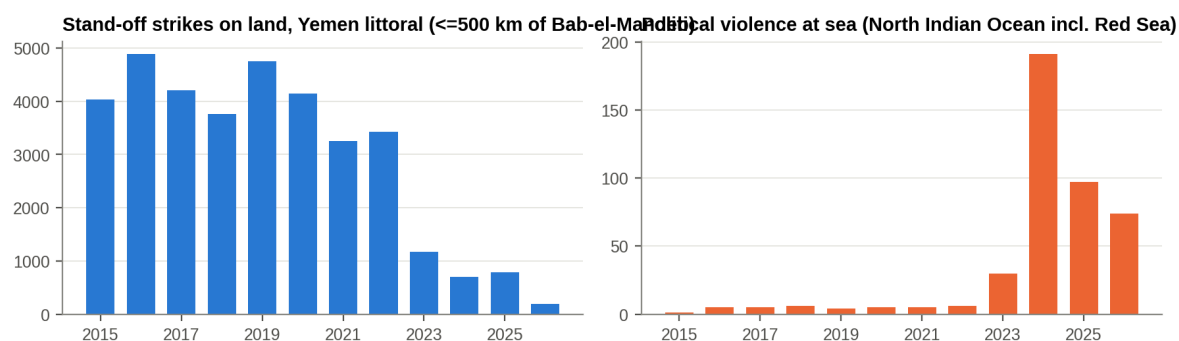
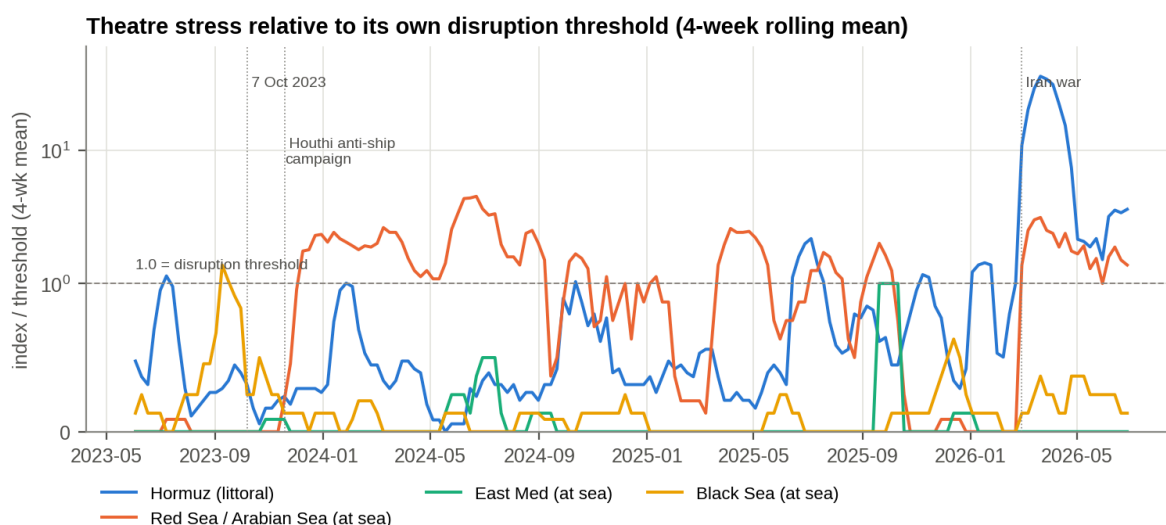


Figure 6.1 : Chokepoint Conflict Intensity Index by theatre, 2015-2026

Red Sea theatre - the violence moved offshore after 2023**Figure 6.2 :** Red Sea theatre: stand-off strikes on the Yemeni littoral versus violence at sea

The Red Sea threat moved offshore. Stand-off strikes on land within 500 km of Bab-el-Mandeb fell from 4,757 in 2019 to 791 in 2025 as Yemen's civil war froze. Over the same years violence at sea rose from 6 events in 2022 to 191 in 2024. An index built on land violence alone would have shown the Red Sea getting *safer* just as it became the most dangerous waterway in the world for merchant shipping. During the 2026 war the Hormuz littoral index averaged **188 times** its 2015-2022 baseline, and the Red Sea at-sea index averaged 46 times its baseline from mid-November 2023 to December 2024.

**Figure 6.3 :** Theatre stress relative to its disruption threshold (4-week rolling mean, symmetric-log scale)**6.2 Forecasting the Hormuz Index****Table 6.1 :** 12-week hold-out back-test of forecasting models (Hormuz littoral index)

Model	MAE (weighted events/wk)	RMSE
Naive (last value)	112	112
Holt damped-trend ETS	92.80	93.30
ARIMA(1,0,1)	56.90	60

ARIMA(1,0,1) on the log-transformed index roughly halved the back-test error of the naive forecast and beat damped-trend exponential smoothing. It was refitted on all data and projected 13 weeks ahead, to the end of September 2026.

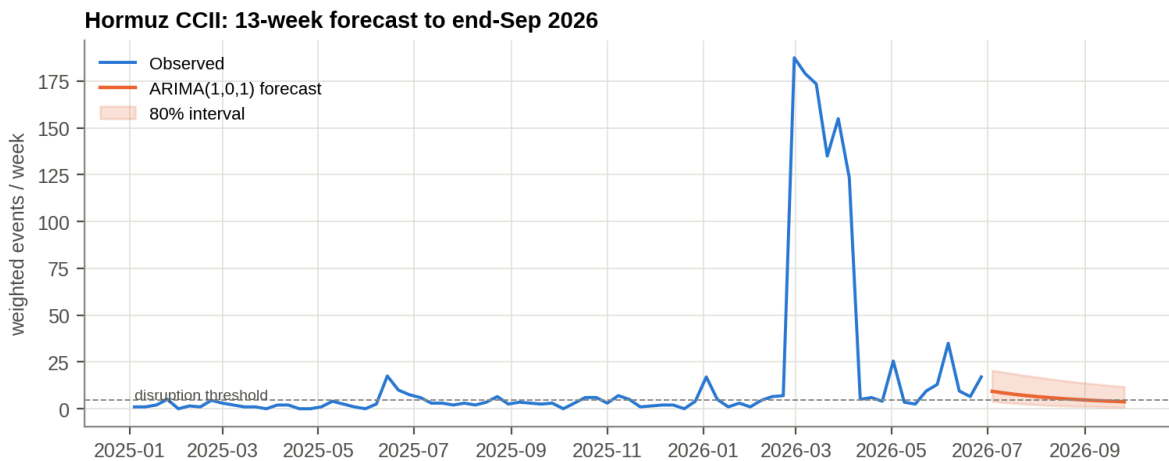


Figure 6.4 : Hormuz CCII: observed values and 13-week ARIMA forecast with 80 per cent interval

The central forecast falls back towards 3.7 weighted events a week by the end of September (80 per cent interval 0.8-11.8), below the disruption threshold of 4.7. However, in 4,000 simulated paths from the fitted model, **99%** cross the threshold in at least one of the 13 weeks. The Hormuz littoral has not returned to its pre-war calm. Flare-ups should be treated as the base case through the third quarter of 2026, not as a tail risk.

6.3 How Long Do Disruptions Last? Survival Analysis

Applying the threshold rule to each theatre produced 32 disruption episodes. One of them, on the Hormuz littoral, was still under way at the end of the data and is right-censored.

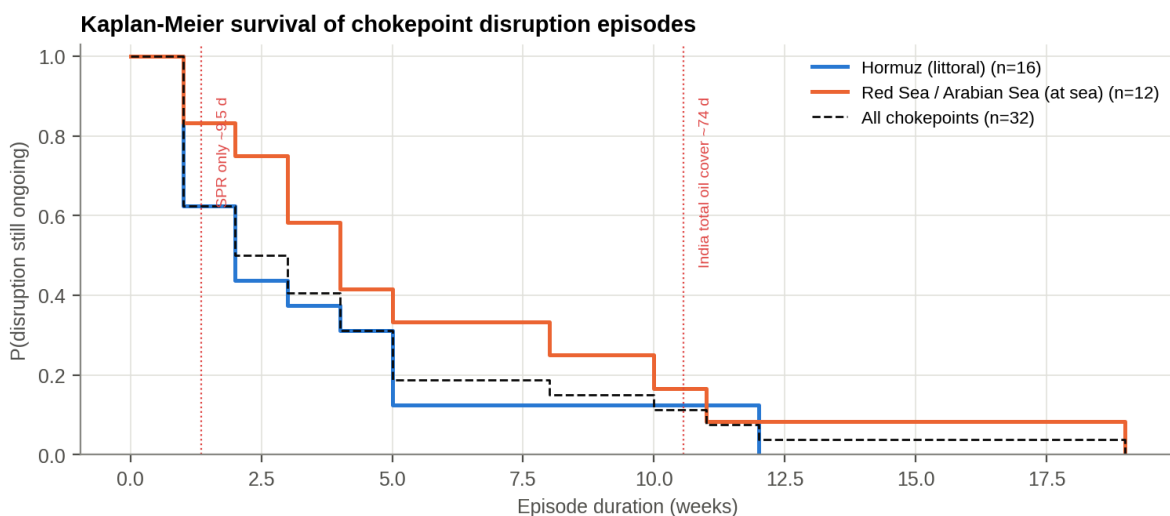


Figure 6.5 : Kaplan-Meier survival of disruption episodes; red lines mark India's strategic oil cover

Table 6.2 : Disruption episodes by theatre

Chokepoint	Episodes	Censored (ongoing)	Median weeks	Longest weeks
Hormuz (littoral)	16	1	2	12
Red Sea / Arabian Sea (at sea)	12	0	4	19
East Med (at sea)	2	0	1	1
Black Sea (at sea)	2	0	1	1

Across the 32 episodes, the median disruption lasts 2.5 weeks, the mean 4.0 weeks and the longest 19 weeks, so the distribution is heavy-tailed. The probability that an episode outlasts **9.5 days**, India's strategic petroleum reserve cover, is **62%**. The probability that it outlasts **74 days**, the total national cover including refiners' stocks, is **11%**. Red Sea episodes last longer than Hormuz episodes (median 4.0 against 2.0 weeks; log-rank $\chi^2 = 8.52$, $p = 0.036$).

Table 6.3 : Cox proportional-hazards model of the hazard of a disruption ending (reference: Black Sea)

Covariate	Coefficient	HR = exp(coef)	p-value	HR 95% CI
Theatre: East Med (at sea)	0.85	2.33	0.46	0.25 - 21.63
Theatre: Hormuz (littoral)	-0.66	0.52	0.43	0.10 - 2.64
Theatre: Red Sea / Arabian Sea (at sea)	-0.77	0.47	0.42	0.07 - 3.03
Episode began after Oct 2023	-0.85	0.43	0.12	0.15 - 1.23

A hazard ratio below 1 means an episode is *less likely to end*, i.e. it lasts longer. Episodes that began after October 2023 have a hazard ratio of 0.43: they are more persistent, though with only 32 episodes the estimate is not significant at the 5 per cent level ($p = 0.12$). **H6 is supported**: disruptions are heavy-tailed and most of them outlast India's strategic reserve.

7. MARITIME TRAFFIC AND AIS DARK ZONES

This chapter analyses 106,533 Sentinel-1 vessel detections from 1-14 March 2026, the first two weeks of the war, and tests hypotheses H3 and H4.

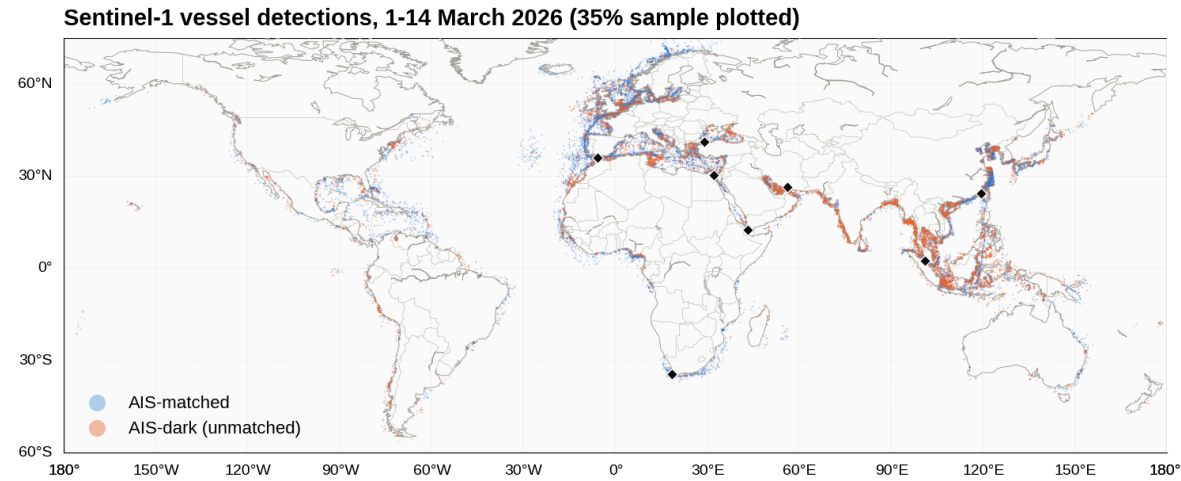


Figure 7.1 : Sentinel-1 vessel detections, 1-14 March 2026 (blue = AIS-matched, orange = AIS-dark)

7.1 Where Are Ships Dark? (H3)

AIS-dark share by sea region - orange: war zones, violet: India's seas

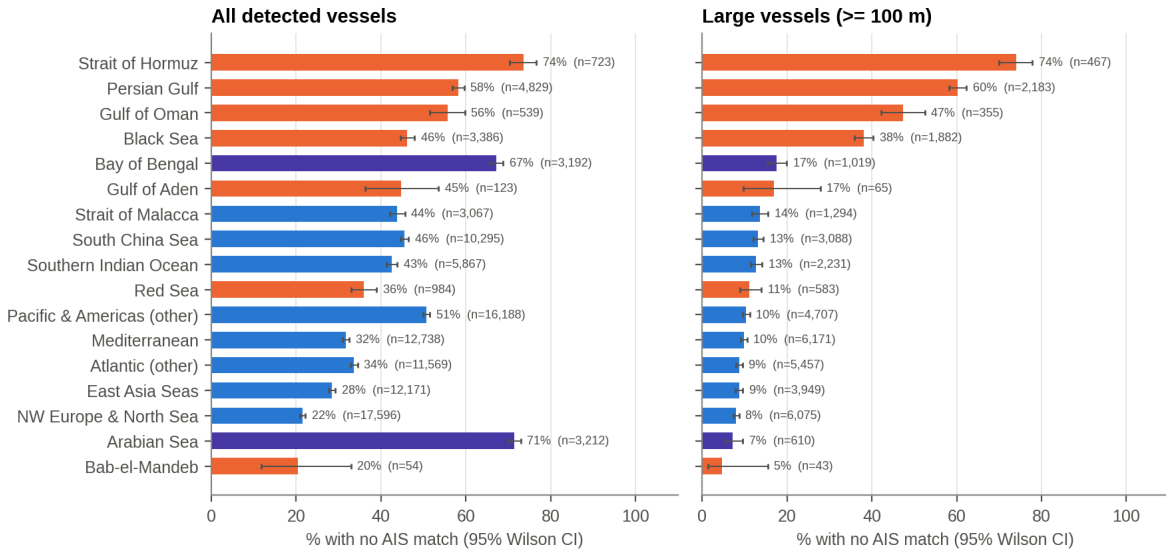


Figure 7.2 : AIS-dark share by sea region, all vessels and large vessels, with 95 per cent Wilson intervals

Table 7.1 : AIS-dark share by region (large = hull length ≥ 100 m)

Region	Detections	Dark share (all)	Large vessels	Dark share (large)	CI low	CI high
Strait of Hormuz	723	73.6%	467	74.1%	69.9%	77.9%
Persian Gulf	4,829	58.2%	2,183	60.2%	58.2%	62.3%
Gulf of Oman	539	55.7%	355	47.3%	42.2%	52.5%
Black Sea	3,386	46.2%	1,882	38.1%	35.9%	40.3%
Bay of Bengal	3,192	67.2%	1,019	17.5%	15.3%	19.9%
Gulf of Aden	123	44.7%	65	16.9%	9.7%	27.8%
Strait of Malacca	3,067	43.9%	1,294	13.5%	11.8%	15.5%
South China Sea	10,295	45.6%	3,088	13.2%	12.0%	14.4%
Southern Indian Ocean	5,867	42.5%	2,231	12.6%	11.3%	14.1%
Red Sea	984	35.9%	583	11.1%	8.8%	14.0%
Pacific & Americas (other)	16,188	50.8%	4,707	10.3%	9.5%	11.2%
Mediterranean	12,738	31.7%	6,171	9.8%	9.1%	10.6%
Atlantic (other)	11,569	33.6%	5,457	8.7%	8.0%	9.4%
East Asia Seas	12,171	28.4%	3,949	8.7%	7.8%	9.6%
NW Europe & North Sea	17,596	21.5%	6,075	8.0%	7.3%	8.7%
Arabian Sea	3,212	71.5%	610	7.2%	5.4%	9.5%
Bab-el-Mandeb	54	20.4%	43	4.7%	1.3%	15.5%

Looking at all ships gives a misleading picture, because small craft are dark almost everywhere. Once the comparison is restricted to **large ships (≥ 100 m)**, which carry AIS by regulation, the war zones stand out: **Strait of Hormuz 74%**, Persian Gulf 60%, Gulf of Oman 47%, Black Sea 38%. Every region at peace lies between about 7 and 17 per cent, and NW Europe is lowest at 8 per cent. Large ships in war zones are dark in 47% of cases against 10% elsewhere: a **relative risk of 4.7** and an **odds ratio of 8.0** ($\chi^2 = 5,101$, $p < 10^{-300}$). **H3 is strongly supported.**

The Black Sea result is independent corroboration. It lies outside the ACLED Middle-East data, and it is the other theatre named in the Datathon problem statement. The Russia-Ukraine war produces the same signature: 38% of large ships dark, about four times the peacetime norm.

The Red Sea shows a different effect: only 11% of large ships are dark, and just 54 detections were made in Bab-el-Mandeb in two weeks. In the Red Sea, ships do not go dark in the war zone. **They stay away.** The Gulf has no route around it, so ships that must load there go dark instead. The two chokepoints show the two responses of shipping to conflict: diversion where a detour exists, blinding where it does not.

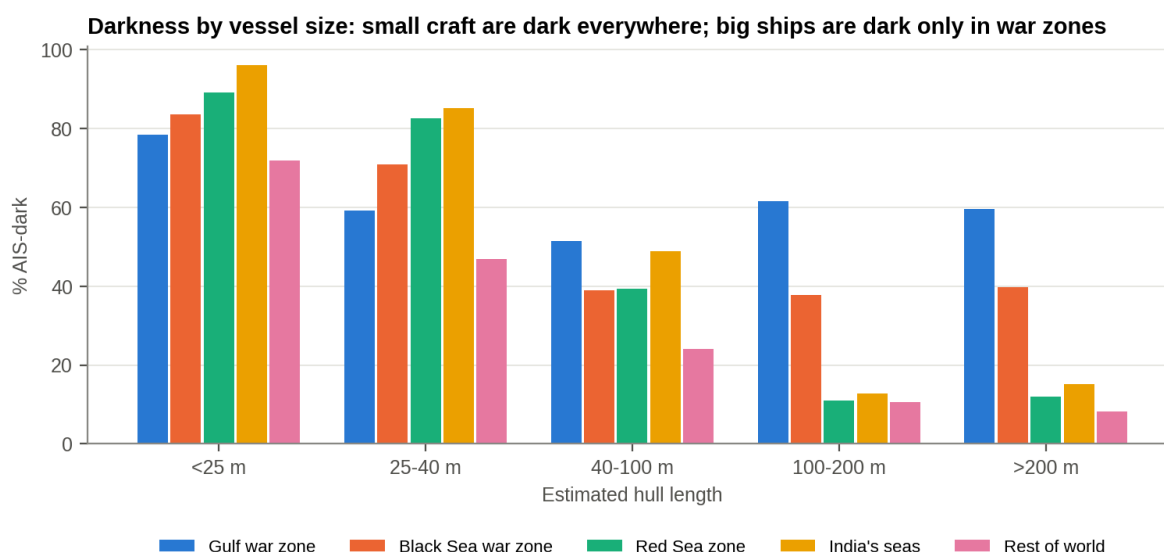


Figure 7.3 : AIS-dark share by vessel size class and zone

In peaceful waters darkness falls steadily with size, from 72 per cent of craft under 25 m to 8 per cent of ships over 200 m. In the Gulf war zone it hardly falls at all: ships over 200 m, which is VLCC and LNG-carrier size, are dark 60 per cent of the time. **In a war zone, size no longer predicts visibility.**

7.2 Statistically Significant Dark Hot Spots (H4)

AIS dark spots in India's western approaches

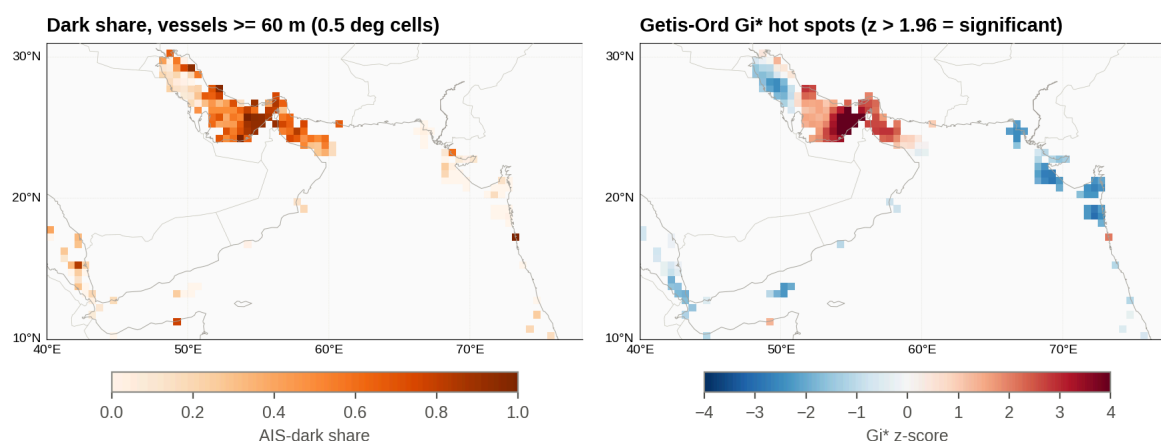


Figure 7.4 : Dark share and Getis-Ord Gi* hot spots, ships ≥ 60 m, India's western approaches (0.5° cells)

Of 183 half-degree cells with enough ships in the Gulf-Arabian Sea window, **43 are significant dark hot spots** ($Gi^* > 1.96$). They form one contiguous belt from Qatar across the UAE coast and Hormuz into the Gulf of Oman. The north-western Gulf (Kuwait, Iraq, the Saudi coast) and India's west coast are **cold spots**: large ships there stay visible.

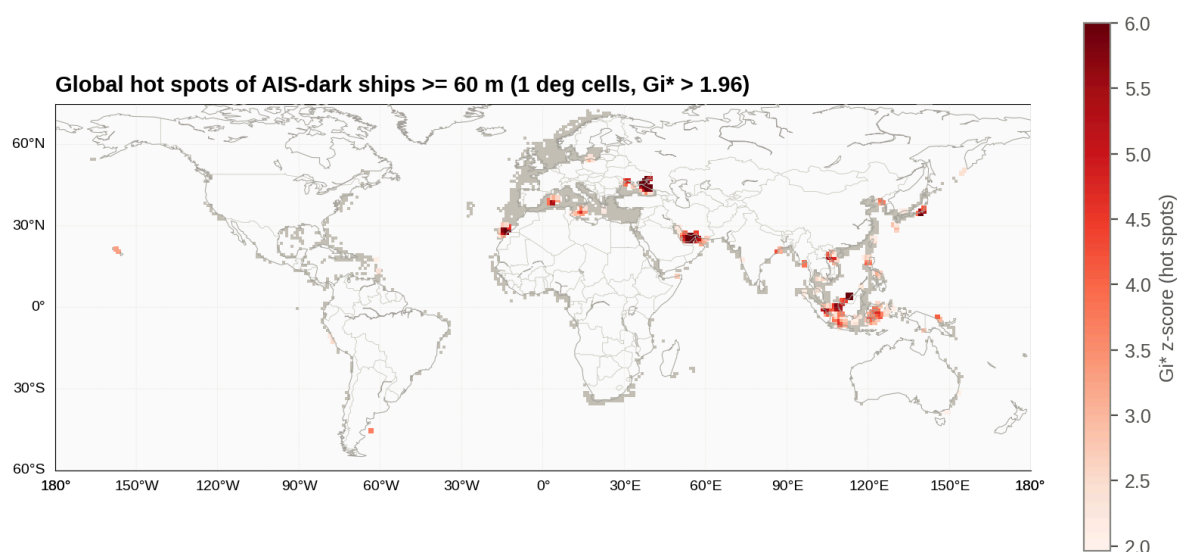


Figure 7.5 : Global G_i^* hot spots of AIS-dark ships ≥ 60 m (1° cells)

Table 7.2 : Significant dark hot-spot cells (1°) by region

Region	Hot-spot cells
Pacific & Americas (other)	54
South China Sea	41
Southern Indian Ocean	35
Mediterranean	22
Persian Gulf	21
Black Sea	20
Atlantic (other)	19
East Asia Seas	18
Strait of Malacca	9
Gulf of Oman	6
NW Europe & North Sea	4
Bay of Bengal	3
Strait of Hormuz	2
Gulf of Aden	1
Arabian Sea	1

Worldwide, 256 of 1,858 one-degree cells are dark hot spots. Relative to the area of each region, the Persian Gulf, Gulf of Oman and Black Sea are the densest. Hot spots elsewhere, in the South China Sea, Southern Indian Ocean and Pacific, mark distant-water fishing grounds where small and medium craft routinely run without AIS [5]. **H4 is supported.**

7.3 Where Do Dark Ships Gather? DBSCAN Clusters

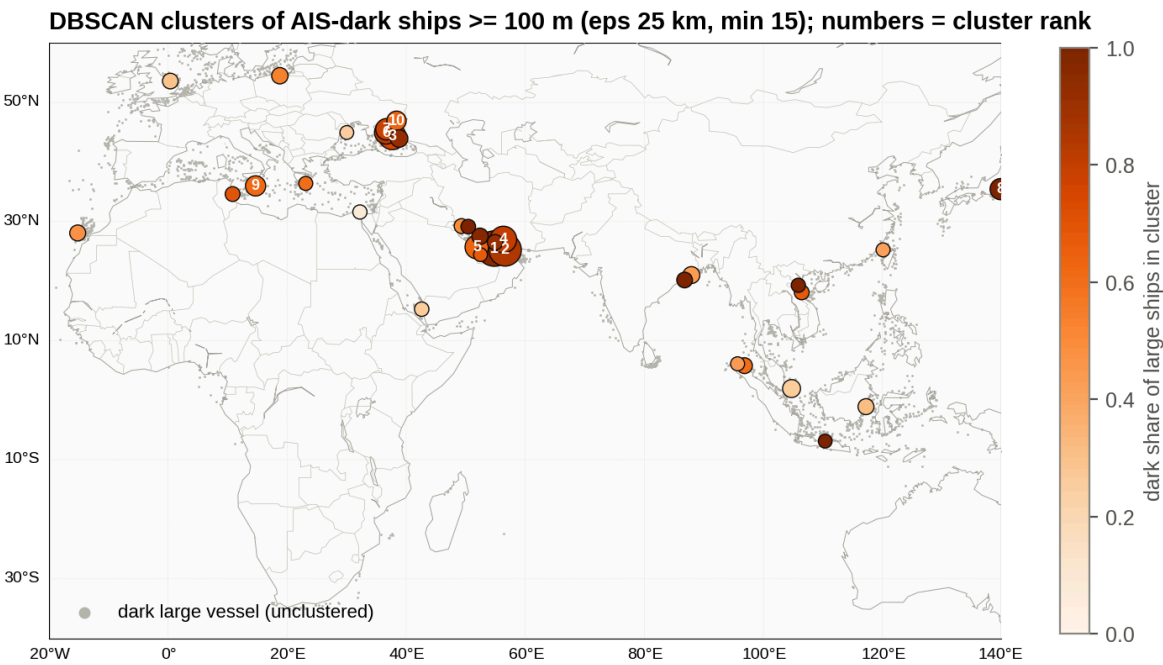


Figure 7.6 : DBSCAN clusters of AIS-dark ships ≥ 100 m (numbers = rank in Table 7.3)

Table 7.3 : Largest clusters of dark ships ≥ 100 m

#	Cluster (nearest reference)	region	Dark large vessels	Dark share of large ships within 30 km	Median length (m)	Days observed
1	Dubai / Jebel Ali	Persian Gulf	660	97%	184	6
2	Fujairah / Khor Fakkan anchorage	Gulf of Oman	462	85%	171	3
3	Novorossiysk / Kerch (Russia)	Black Sea	226	75%	174	5
4	Strait of Hormuz (Bandar Abbas-Qeshm)	Strait of Hormuz	207	80%	179	3
5	Ras Laffan / Doha (Qatar)	Persian Gulf	126	64%	174	5
6	Novorossiysk / Kerch (Russia)	Black Sea	122	83%	149	4
7	Novorossiysk / Kerch (Russia)	Black Sea	88	78%	123	4
8	35.5N 139.9E	East Asia Seas	84	100%	127	1
9	Malta	Mediterranean	69	62%	167	3
10	Novorossiysk / Kerch (Russia)	Black Sea	58	64%	129	5
11	Singapore Strait / EOPL Johor	South China Sea	43	26%	255	1
12	Novorossiysk / Kerch (Russia)	Black Sea	39	91%	164	4
13	Kolkata / Haldia	Bay of Bengal	34	44%	196	2
14	54.5N 18.8E	NW Europe & North Sea	30	54%	149	3
15	Ras Laffan / Doha (Qatar)	Persian Gulf	30	97%	175	4

DBSCAN found 36 clusters. The two largest are off **Dubai/Jebel Ali (660 dark large ships; 97% of all large ships there dark)** and the **Fujairah/Khor Fakkan anchorage**, the main bunkering and waiting area outside Hormuz. Together with the Hormuz and Ras Laffan clusters, they are fleets of tankers and gas carriers waiting at anchor with AIS off. That is the physical form of a supply chain on hold. The Novorossiysk/Kerch clusters are the Black Sea equivalent. The Kolkata/Haldia cluster (44 per cent dark) is the only large-ship cluster in India's own waters and is flagged for follow-up. The single-day Tokyo Bay cluster is most likely an AIS-matching gap and is treated as an artefact.

7.4 The Daily Picture in the Gulf

Daily picture in the Gulf, 1-14 March 2026 (weeks 1-2 of the war)

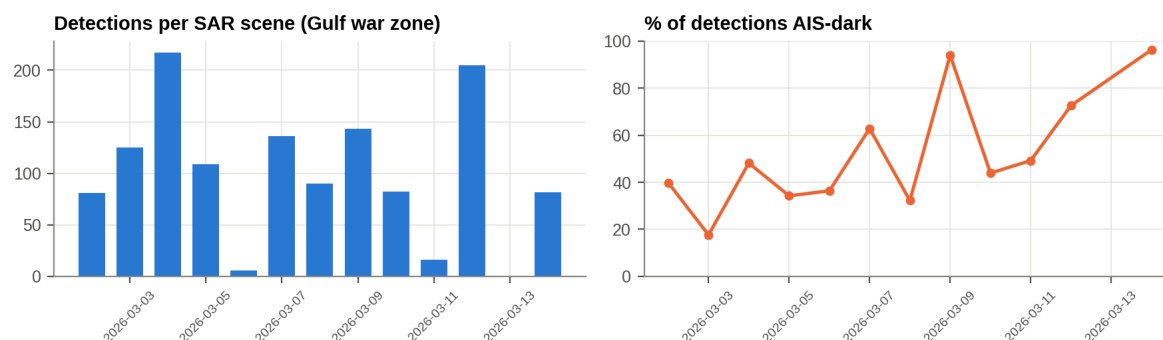


Figure 7.7 : Gulf war zone: detections per SAR scene and share AIS-dark, 1-14 March 2026

Over the first two weeks of the war, darkness in the Gulf **rose** from 35% of detections in the first three imaged days to 73% in the last three (Spearman $\rho = 0.68$, $p = 0.015$). Two days (6 and 12 March) had very few detections per scene, and their shares should be read with caution. The trend holds with them removed.

7.5 Who Is Sailing? Flag States of Visible Ships

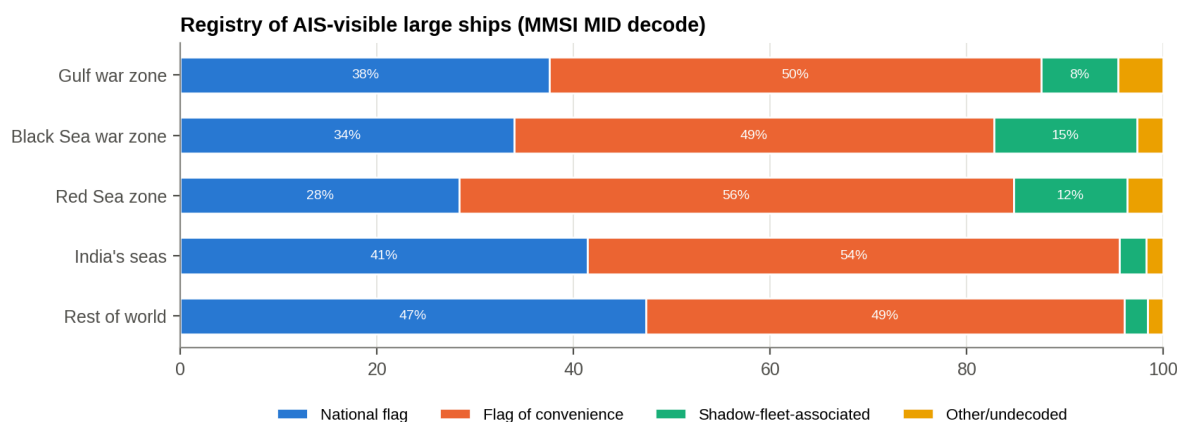


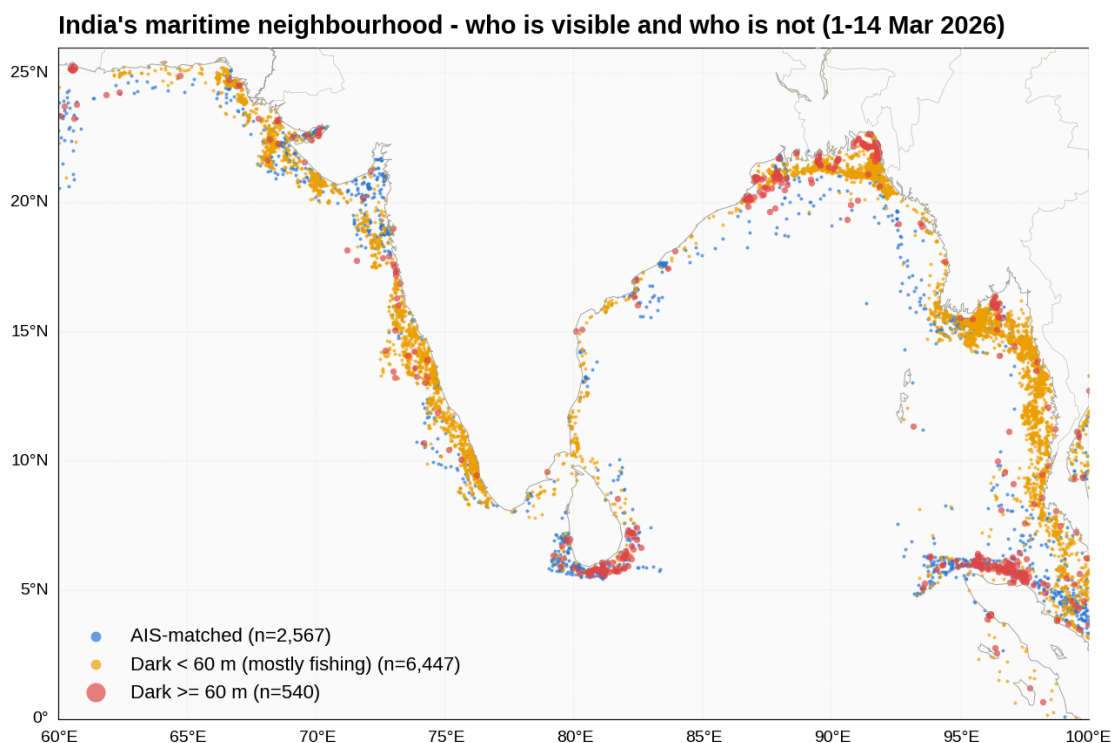
Figure 7.8 : Registry class of AIS-visible large ships by zone (MMSI MID decode)

Table 7.4 : Top flag states of visible large ships, Gulf war zone against rest of world

Flag	Gulf war zone %	Rest of world %
Panama	17.40	11.70
Liberia	9.90	12.80
Marshall Islands	9.20	9.50
Iran	6.90	0.10
UAE	6	0
Other/undecoded	4.60	1.60
Singapore	3.70	5
Comoros	3.60	0.20
Malta	3.20	5.70
Saudi Arabia	3.20	0.10

Flags of convenience make up about half of visible large ships everywhere, so they do not distinguish the war zones. What does distinguish them is **shadow-fleet-associated registries** (Gabon, Cameroon, Comoros, Sierra Leone, Togo, Cook Islands, Palau and similar): 7.8 per cent of visible large ships in the Gulf and 14.6 per cent in the Black Sea, against 2.4 per cent elsewhere. Iranian-flagged ships make up 6.9 per cent of visible large ships in the Gulf against 0.1 per cent worldwide. The ships still transmitting in the war zone lean towards registries with weak oversight, which adds to the identity problem.

7.6 India's Maritime Neighbourhood

**Figure 7.9 :** Arabian Sea and Bay of Bengal: AIS-matched, small dark and large dark detections

India's seas have the highest overall dark shares in the dataset (Arabian Sea 72%, Bay of Bengal 67%). This is **not** a conflict signal: 94% of craft under 40 m are dark, and 85% of dark detections score as fishing vessels. Large ships off India are visible (Arabian Sea 7% dark, Bay of Bengal 18%). The security point is that India's near seas contain thousands of small, uncooperative craft, the same class of craft used in the 26/11 Mumbai attack. The domestic maritime picture depends on non-AIS sensing.

7.7 Identity Integrity

Table 7.5 : Indicators of AIS identity anomalies

Indicator	Global	Gulf war zone
Malformed MMSI (MID outside 2xx-7xx)	1,565	17
Placeholder MMSI (xxx000000)	94	6
Physically impossible re-sighting (>50 kn over >20 km)	15	1
'noisy_vessel' AIS tracks	2,187	73

Beyond darkness, the data show 1,565 malformed MMSIs, 94 placeholder identities (e.g. 2000000000, 4120000000) and 15 cases of one MMSI appearing at two places that would need speeds above 50 knots, i.e. cloned or spoofed identities. These counts are small but not negligible, and they show that an AIS match alone does not confirm a ship's identity.

8. PREDICTING AIS DARK SPOTS

The Datathon problem statement asks participants to *predict* AIS dark spots. A model was trained on 62,624 detections of ships ≥ 40 m (the AIS-carrying fleet; base dark rate 19.6%). It uses five features that describe the ship and its conflict geography, and it deliberately excludes latitude and longitude so that it cannot simply memorise where the dark ships are.

Table 8.1 : Logistic regression: odds ratios per 1 SD of each feature (all 62,624 ships ≥ 40 m)

Feature	Coef (per 1 SD)	Odds ratio	OR 95% CI	p-value
Hull length (log m)	-0.63	0.53	0.52 - 0.55	<0.001
Fishing-vessel score	-0.21	0.81	0.78 - 0.83	<0.001
Distance to nearest conflict ADMIN1 (log km)	-0.50	0.60	0.58 - 0.63	<0.001
Conflict events within 500 km (log)	-0.03	0.97	0.94 - 1.01	0.092
Distance to chokepoint (log km)	0.01	1.00	0.98 - 1.03	0.703

Holding size and type constant, each 1-SD increase in (log) distance from the nearest active conflict province lowers the odds of darkness to 0.60 times. The raw event count within 500 km adds nothing once distance is known. *Proximity* to the fighting matters; the *volume* of fighting does not.

Table 8.2 : Model skill, leave-one-region-out cross-validation

Model	ROC-AUC	PR-AUC	Brier	F1 @0.5	Accuracy
Logistic regression	0.62	0.28	0.15	0.09	0.80
Gradient-boosted trees	0.69	0.37	0.14	0.20	0.81
Baseline (always 'not dark')	0.50	0.20	0.16	0	0.80
GBT, random 5-fold CV (leaky, for contrast)	0.74	0.48	0.13	0.32	0.82

Predicting AIS darkness from conflict geography

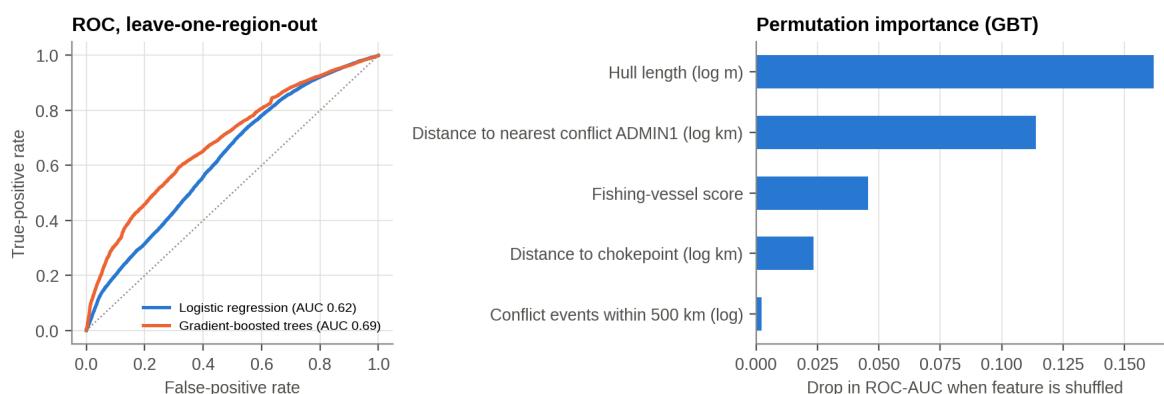


Figure 8.1 : ROC curves (leave-one-region-out) and permutation importance

Evaluated only on sea regions it had never seen, the gradient-boosted model reaches **ROC-AUC 0.69** and **PR-AUC 0.37**, almost twice the 0.20 a random model would score. Ordinary

random cross-validation would have reported AUC 0.74. That inflation from spatial leakage is exactly what Roberts et al. warn against [14], and the lower, honest figure is the one relied on here. **H5 is supported:** conflict geography carries real, transferable information about where ships will go dark.

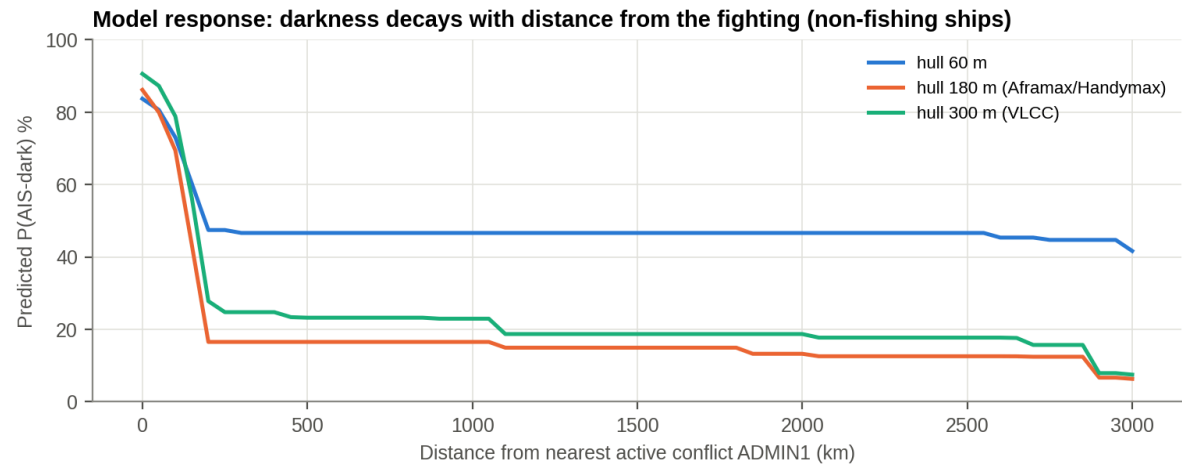


Figure 8.2 : Model response: predicted darkness against distance from active conflict

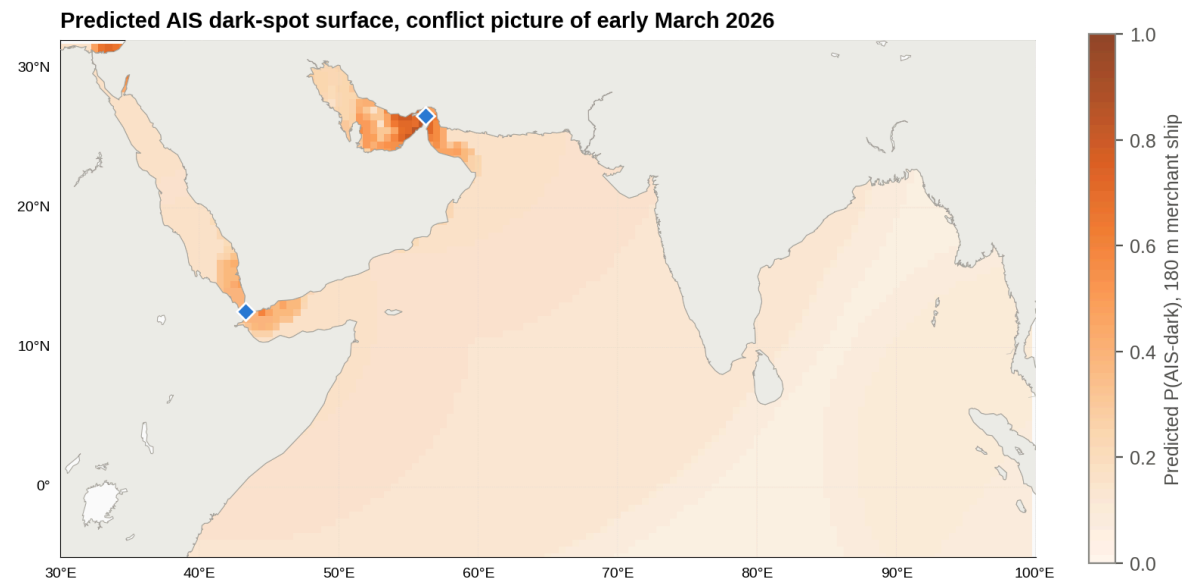


Figure 8.3 : Predicted AIS dark-spot surface for a 180 m merchant ship, early-March-2026 conflict picture

Table 8.3 : Predicted probability that a 180 m merchant ship is AIS-dark at points of interest to India

Location	Predicted P(dark), 180 m ship
Strait of Hormuz	86%
Fujairah anchorage	60%
Gulf of Aden	32%
Ras Tanura	25%
Chabahar	17%
Kandla/Mundra approaches	15%
Mumbai approaches	13%
Kochi approaches	12%
Colombo	7%

The model's surface reproduces the observed belt of darkness along Hormuz and the UAE coast and a secondary area around Bab-el-Mandeb and the Gulf of Aden. It also gives a *transferable* rule. Given a new conflict picture, such as a future escalation near Chabahar or the Makran coast, the same model can be re-run to predict where the cooperative maritime picture will fail before any satellite pass confirms it.

9. KEY FINDINGS AND IMPACT ASSESSMENT

9.1 Synthesis of Findings

2.9x weekly political violence in the 2026 war regime vs the prior year	124x rise in violence inside GCC energy states	74% large ships AIS-dark in the Strait of Hormuz	4.7x relative risk of darkness for large ships in war zones	62% disruptions outlasting India's 9.5-day SPR
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Table 9.1 : Key findings of the study

#	Finding	Section
F1	Violence rose 2.9-fold in the 2026 war regime; PELT dated the regime shifts to 7 Oct 2023 and 28 Feb 2026 without being given the dates.	5.1
F2	84% of war-period violence was stand-off strikes (air, drone, missile, artillery), a sharp jump that marks the onset of inter-state war.	5.2
F3	The war reached the energy coast: GCC violence rose 124-fold, the Hormuz littoral index 188-fold.	5.2, 6.1
F4	The Red Sea threat moved offshore: land strikes fell ~6-fold while at-sea violence rose ~30-fold; a land-based index would miss it.	6.1
F5	Disruptions are heavy-tailed: median 2.5 wk, max 19 wk; 62% outlast the SPR, 11% outlast 74 days; Hormuz stays elevated into Q3-2026 (99% of simulated paths cross the threshold).	6.2, 6.3
F6	Large ships in war zones are 4.7x more likely to be AIS-dark (Hormuz 74%, Black Sea 38%); darkness in the Gulf rose over the first two weeks of the war.	7.1, 7.4
F7	Two responses to conflict: blinding where there is no detour (Gulf) and diversion where there is (Red Sea).	7.1
F8	Dark ships gather at the Dubai/Jebel Ali and Fujairah anchorages: the supply chain is held at anchor with AIS off; shadow-fleet registries are over-represented about 3x.	7.3, 7.5
F9	India's near seas are dominated by thousands of small, non-AIS craft; large ships off India remain visible.	7.6
F10	Proximity to conflict predicts darkness in unseen regions (ROC-AUC 0.69); AIS identity itself is unreliable (clones, placeholders, malformed IDs).	7.7, 8

9.2 Impact on the Globe

- **Energy security.** About a fifth of the world's oil liquids and a large share of its LNG pass through Hormuz [2]. F3 and F5 show the strait's shoreline became an active front and stayed above its disruption threshold for most of February to June 2026. The world energy market now has to price chokepoint risk as a recurring condition, not a rare event.
- **Freight, time and cost.** F4 and F7 show that where a detour exists, as in the Red Sea, conflict produces diversion. Cape routing adds 10-14 days to Asia-Europe voyages, absorbs fleet capacity and raises freight and insurance costs worldwide [1]. The survival curves (F5) show these episodes last weeks to months, not days.

- **Loss of the common maritime picture.** F6 and F10 show that in war zones the cooperative identity system on which collision avoidance, search and rescue, sanctions enforcement, insurance and port-state control depend fails for most large ships. Navigation safety deteriorates, and the chance of misidentifying and striking a neutral ship rises.
- **Sanctions and the shadow fleet.** F8 shows dark ships clustering at the Gulf and Black Sea anchorages and shadow-fleet registries about three times over-represented. The informal, poorly insured fleet grows exactly where oversight is weakest, raising the risk of environmental disasters and unattributable incidents.
- **Contagion of stand-off warfare.** F2 shows that cheap drones and missiles have become the main tool of regional war. Any littoral actor can now hold a chokepoint at risk from the shore.

9.3 Impact on India

- **Energy import exposure.** India imports close to nine-tenths of its crude oil. Open-source estimates put a large share of that crude (commonly 40-50 per cent), most of its LPG and much of its LNG as coming from the Gulf through Hormuz [2, 19]. The heavy-tailed disruption curve (F5) means that about 62% of disruptions would outlast the strategic petroleum reserve (~9.5 days) and about 11% would outlast the total national cover of about 74 days [19].
- **Trade routes to Europe, Africa and the US East Coast.** The Red Sea diversion (F4, F7) lengthens India's westbound container and product-tanker routes and raises their cost, eroding the competitiveness of exports such as refined products, engineering goods, textiles and pharmaceuticals. The Houthi campaign episodes lasted a median of four weeks and up to 19 weeks.
- **Connectivity projects.** The International North-South Transport Corridor through Chabahar and the India-Middle East-Europe Economic Corridor (IMEC) both pass through the theatres analysed. The predicted dark-risk at Chabahar (Table 8.3) and the violence in Sistan-Baluchestan and Hormozgan (Chapter 5) point to delays and higher risk premia for both.
- **Diaspora and remittances.** About nine million Indians live in the GCC, which sends India a large share of its remittances [20]. The 124-fold rise in violence inside the GCC (F3) makes large-scale non-combatant evacuation a planning case, not a contingency.
- **Maritime domain awareness at home.** F9 shows that India's own seas are dominated by thousands of small, non-AIS craft. F10 shows that even AIS-visible identities can be cloned or spoofed. The national maritime picture cannot rely on AIS alone.

9.4 Impact on the Indian Army

The Army is not a maritime force, but its war-fighting ability depends on the same supply chains, and it faces the same technology shift that this data records. The findings bear on the Army in six ways.

- **POL and war-wastage reserves.** Mechanised formations, Army Aviation and the logistics fleet run on imported crude refined in India. A chokepoint disruption that outlasts national reserves (F5) would lead to rationing, in which defence consumption competes with the civil economy. The prescriptive model below turns F5 into a stock-holding decision.
- **Imported equipment, spares and ammunition.** Sea-lifted defence imports and the spares for imported platforms move on the same routes. Longer, more variable lead-times (F4, F5, F7) mean that re-order points and safety stocks based on peacetime lead-times will be too low.
- **The stand-off threat to logistics nodes.** F2 shows that drones and missiles make up most of the violence in modern regional war and that GCC energy infrastructure far from any front line was struck (F3). Army depots, railheads, POL points and forward airfields need the same layered air defence, counter-UAS, dispersal and hardening as the Gulf's terminals needed.
- **Navigation warfare and identity.** The AIS darkness and identity anomalies (F6, F10) are the maritime form of electronic warfare that also denies and spoofs GNSS on land. Army drones, loitering munitions, precision artillery and timing networks need NavIC/multi-constellation, anti-jam and GNSS-denied fallback capability.
- **Western front and hybrid spill-over.** Violence in Sistan-Baluchestan (among Iran's most violent provinces over the decade), the Makran littoral and a weakened Iranian periphery all border Pakistan's Balochistan. These affect the western-front threat picture and the security of connectivity projects.
- **Out-of-area and HADR roles.** Large-scale evacuation of the Gulf diaspora (Section 9.3) would draw on Army transit, medical and staging capacity in support of the Navy and Air Force. Earlier evacuations (Op Ganga 2022, Op Kaveri 2023, Op Ajay 2023) are the reference cases.

Prescriptive stock-cover curve from observed disruption durations

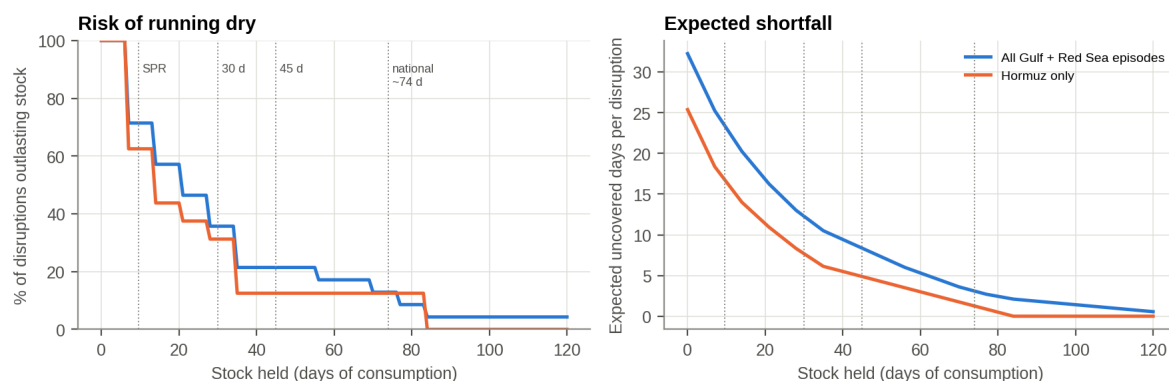


Figure 9.1 : Prescriptive stock-cover curve from observed Gulf and Red Sea disruption durations

Table 9.2 : Decision table: stock held against disruption risk and expected shortfall

Stock (days)	P(disruption outlasts stock) %	Expected uncovered days	Shortfall avoided vs zero stock %
0	100	32.20	0
10	71.40	23.10	28.30
15	57.10	19.70	39
30	35.70	12.30	61.90
45	21.40	8.40	74.10
60	17.10	5.30	83.50
74	12.90	3.10	90.40
90	4.30	1.80	94.30

With no stock, an average Gulf or Red Sea disruption leaves 32 uncovered days. Holding **30 days** of stock cuts the expected shortfall to 12.3 days, and **45 days** to 8.4 days, avoiding 74 per cent of the shortfall. The returns flatten at about **35 days**: beyond this point each additional day of stock removes less than a quarter of a day of expected shortfall. The same curve can be applied to any disruption-sensitive commodity (POL, lubricants, critical imported spares) to set holdings on evidence rather than precedent.

9.5 Limitations

- The SAR data cover only 14 days from a single satellite (Sentinel-1A), so the pre-war level of darkness in the same waters cannot be observed directly. Peacetime regions and the Black Sea serve as the comparison groups instead.
- 'Unmatched' is not always deliberate: GNSS spoofing, AIS reception gaps and matching-algorithm limits also produce darkness. In the Gulf these are themselves effects of the war.
- ACLED aggregates are at ADMIN1 resolution, and its at-sea events carry a nominal basin centroid. Distances to conflict are therefore approximate.

- The survival analysis rests on 32 episodes, so Cox coefficients have wide intervals. MID-to-flag decoding and registry classes are approximations, and registry class is not proof of any particular ship's conduct.
- Figures quoted from secondary sources (oil import shares, reserve days, diaspora) are indicative and cited, and they were not derived from the Datathon datasets.

10. RECOMMENDATIONS

Each recommendation is traced to the finding(s) that support it (Table 9.1).

Table 10.1 : Recommendations traced to findings

Level	#	Recommendation	Based on
National (strategic)	R1	Expand strategic crude and LPG reserves and set national cover from the disruption-survival curve rather than a fixed number of days, with a target at or beyond the ~35-45-day point where returns flatten, for the Hormuz-dependent share of imports.	F5
National (strategic)	R2	Diversify energy sources and routes: more non-Gulf crude and LNG, west-coast refinery access to Red Sea-bypass pipelines (Yanbu, Fujairah), and long-term charters that include war-risk clauses.	F3, F5, F7
National (strategic)	R3	Build a national Chokepoint Early-Warning System that tracks the CCII, the stand-off strike share and the large-ship dark share weekly, with the thresholds from this study as triggers.	F1, F2, F6
Maritime / MDA	R4	Fuse SAR (NISAR, EOS-04/RISAT, commercial SAR) with AIS at IFC-IOR and flag every dark large ship automatically, prioritising the hot-spot belt and the predicted dark-risk surface.	F6, F8, F10
Maritime / MDA	R5	Treat AIS identity as untrusted: build automated checks for impossible re-sightings, placeholder and malformed MMSIs and shadow-fleet registries, and share them with ports and insurers.	F8, F10
Maritime / MDA	R6	Extend coverage of small craft in India's seas (fishing-vessel transponders, coastal radar, SAR) so that the non-AIS population is known.	F9
Indian Army	R7	Re-baseline POL, lubricant and critical-spares War Wastage Reserves using the expected-shortfall method (Eq. 3.6), and re-order points using lead-time distributions measured during crises.	F4, F5
Indian Army	R8	Give priority to layered air defence, counter-UAS, dispersal and hardening of depots, POL points and railheads, because stand-off strikes now dominate regional war.	F2, F3
Indian Army	R9	Make GNSS resilience mandatory for drones, loitering munitions, precision fires and timing: NavIC plus multi-GNSS, anti-jam antennas, inertial or vision fallback, and field training under jamming and spoofing.	F6, F10
Indian Army	R10	Set up an Army Data Analytics Cell (DG Log / DGMO / DGIS) that runs this pipeline and its Power BI dashboard as a standing supply-chain and conflict early-warning product.	All
Tri-service	R11	Pre-plan non-combatant evacuation from the GCC as a live case, with Army transit, medical and staging components rehearsed with the Navy and IAF.	F3
Tri-service	R12	Use Indigenisation (Aatmanirbharta) priority lists weighted by route exposure, so that items that move through Hormuz or the Red Sea and have long lead-times are indigenised or dual-sourced first.	F4, F5, F7

11. WAY FORWARD

The recommendations can be implemented in three phases, each building on the data products already produced by this study.

Table 11.1 : Phased way forward

Phase	Actions	Deliverable
Phase 1 (0-6 months)	Institutionalise the pipeline. Host the Power BI dashboard (Appendix B) at Army HQ; run the weekly CCII and stand-off-share refresh from ACLED; set alert thresholds; start the POL and spares WWR review using Eq. 3.6.	Weekly early-warning brief; revised stock norms for trial
Phase 2 (6-18 months)	Fuse sensors and automate. Ingest national SAR (NISAR, EOS-04) and commercial SAR with AIS; run automated dark-ship and identity-anomaly detection; retrain the dark-spot model monthly; add freight-rate, insurance and port-call data to measure physical impact directly.	Near-real-time dark-ship alerts; validated dark-risk maps
Phase 3 (18-36 months)	Move to prescriptive, joint decision support. Integrate with the tri-service logistics and inventory systems; simulate stock, route and evacuation decisions under scenarios; extend the same methods to the South China Sea and Malacca theatres; train staff in data-driven logistics through CDM and the Army's own analytics courses.	Joint supply-chain resilience system; trained analytics cadre

Future research. Several extensions would strengthen the results: (a) extend the SAR window to a pre-war baseline for the same waters, which would give a true difference-in-differences estimate of war-induced darkness; (b) add ship-level AIS gap events and port calls to measure turnaround and diversion directly; (c) model GNSS-interference zones explicitly to separate spoofing from deliberate switch-off; (d) move from ADMIN1 centroids to event-level ACLED coordinates.

12. CONCLUSION

12.1 Conflict Analytics

- a. Middle-East political violence moved through distinct regimes, and change-point detection dated the shifts to 7 October 2023 and 28 February 2026 without being given them. The war regime ran at 2.9 times the preceding year.
- b. The 2026 war was fought mainly with stand-off weapons (84 per cent of violence), reached the GCC energy coast (124-fold rise) and put the Hormuz littoral at 188 times its baseline.
- c. In the Red Sea the threat moved offshore. Chokepoint risk must be measured at sea, not only on land.

12.2 Disruption Survival

- a. Disruption episodes are heavy-tailed (median 2.5 weeks, longest 19 weeks). 62% outlast India's strategic reserve and 11% outlast total national cover.
- b. Holding about 35-45 days of stock covers most of the expected shortfall. Beyond about 35 days each extra day of stock adds little protection.

12.3 AIS Dark Zones

- a. Large ships were AIS-dark in 74% of cases in the Strait of Hormuz and 38% in the Black Sea, against about 10 per cent in peacetime waters (relative risk 4.7). The identification crisis named in the problem statement is real and measurable.
- b. Conflict produces blinding where there is no detour and diversion where there is. Dark tankers gathered at the Gulf anchorages, and shadow-fleet registries were over-represented.
- c. Proximity to conflict predicts darkness in sea regions the model never saw (ROC-AUC 0.69), which makes it possible to forecast where the maritime picture will fail.

Closing statement. The first casualty of war at a chokepoint is *visibility*. Physical disruption follows it, lasts longer than intuition suggests, and reaches India's energy, trade, diaspora and military logistics. The same data and methods that measured this can warn of it. By turning the pipeline, index, dark-spot model and stock-cover curve built here into standing tools, India and the Indian Army can base these decisions on data rather than on precedent, which is the aim of the Datathon.

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APPENDIX A - TOOLCHAIN AND REPRODUCIBILITY

The analysis is fully scripted. The repository folder *datathon2026/* contains the pipeline below. Running *python analysis/run_all.py* rebuilds every artefact from the two raw files in about one minute on a laptop.

datathon2026/ analysis/common.py helpers	shared styling, regions, chokepoints, MMSI-MID registry,
analysis/01_preprocess.py spatial join	cleaning, validation, feature engineering, conflict
analysis/02_conflict.py GCC spill-over, maps	Chapter 5: trends, event mix, heat map, PELT regimes,
analysis/03_chokepoint_index.py forecast, Kaplan-Meier, Cox	Chapter 6: CCII, thresholds, ARIMA back-test and
analysis/04_sar.py DBSCAN, flags, identity	Chapter 7: dark shares, chi-square, Gi* hot spots,
analysis/05_model.py risk surface	Chapter 8: logit, gradient-boosted trees, spatial CV,
analysis/06_decision.py analysis/07_powerbi_export.py	Section 9.4: expected-shortfall stock-cover model star-schema tables for the Power BI dashboard
figures/ tables/ powerbi/ data/metrics.json report/build_report.py	

Table A.1 : Software stack

Layer	Tools
Language / runtime	Python 3.11
Data manipulation	pandas 2.x, NumPy, PyArrow (Parquet)
Statistics	SciPy (Mann-Whitney, chi-square, Spearman), statsmodels (Logit, ARIMA/SARIMAX, ETS, Kaplan-Meier, Cox PHReg, Wilson CI)
Machine learning	scikit-learn (HistGradientBoosting with monotonic constraints, GroupKFold spatial CV, DBSCAN, BallTree, permutation importance)
Time series	ruptures (PELT change-points)
Visualisation / GIS	Matplotlib, Basemap (coastlines), custom Getis-Ord Gi*
Dashboard	Microsoft Power BI Desktop (.pbix) on the exported star schema
Report	HTML/CSS rendered to PDF with headless Chromium; all numbers injected from metrics.json

APPENDIX B - POWER BI DASHBOARD DATA MODEL

The Datathon requires a Power BI (.pbix) file alongside this report. The pipeline exports a star schema (*powerbi/*) that loads directly into Power BI Desktop. The model, relationships and core DAX measures are given below. The step-by-step build of the four dashboard pages is in *powerbi/POWERBI_BUILD_GUIDE.md*.

Table B.1 : Power BI star schema

Table	Rows	Key columns	Relationship
FactConflictWeekly	149,814	WEEK, ADMIN1_ID, EVENT_TYPE, SUB_EVENT_TYPE, EVENTS, FATALITIES, flags	ADMIN1_ID -> DimAdmin1; WEEK -> DimDate[DATE]
DimAdmin1	243	ADMIN1_ID, COUNTRY, ADMIN1, LAT, LON, GCC, MARITIME, KM_TO_HORMUZ ...	-
FactVesselDetections	106,533	DETECTION_ID, DATE, lat, lon, REGION, ZONE, length_m, AIS_DARK, LARGE_100M, flag, flag_class ...	DATE -> DimDate[DATE]
DimDate	4,383	DATE, YEAR, QUARTER, MONTH, WEEK_START	-
t6_ccii_weekly, t6_2_episodes, t7_4_dark_clusters, t8_3_poi_risk, t9_1_stock_decision	-	analysis outputs	stand-alone

```

Events          = SUM(FactConflictWeekly[EVENTS])
Fatalities      = SUM(FactConflictWeekly[FATALITIES])
Stand-off share = DIVIDE(CALCULATE([Events], FactConflictWeekly[DRONE_MISSILE] = TRUE()),
                        CALCULATE([Events], FactConflictWeekly[POLITICAL_VIOLENCE] = TRUE()))
Detections      = COUNTROWS(FactVesselDetections)
Dark share      = DIVIDE(CALCULATE([Detections], FactVesselDetections[AIS_DARK] = TRUE()), [Detections])
Dark share large = CALCULATE([Dark share], FactVesselDetections[LARGE_100M] = TRUE())
Dark RR vs world = DIVIDE([Dark share large],
                        CALCULATE([Dark share large], FactVesselDetections[ZONE] = "Rest of world"))

```

Dashboard pages. (1) Conflict Pulse: weekly events and fatalities, event-type mix, map of ADMIN1 bubbles, GCC slicer. (2) Chokepoint Watch: CCII lines by theatre with threshold lines, episode table, forecast. (3) Dark Ships: map of detections coloured by AIS_DARK, dark share by region and size, cluster table, flag class. (4) Decision: stock-cover decision table, points-of-interest risk, and a what-if parameter for days of stock.